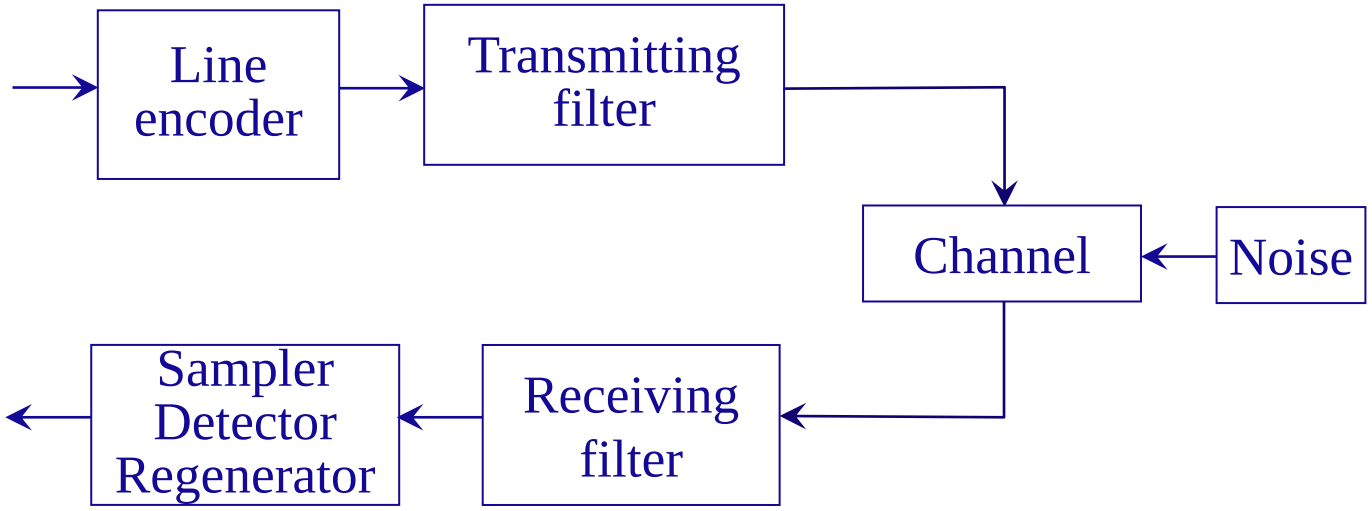
Digital Passband Transmission

Binary Amplitude Shift Keying
Binary Frequency Shift Keying
Binary Phase Shift Keying

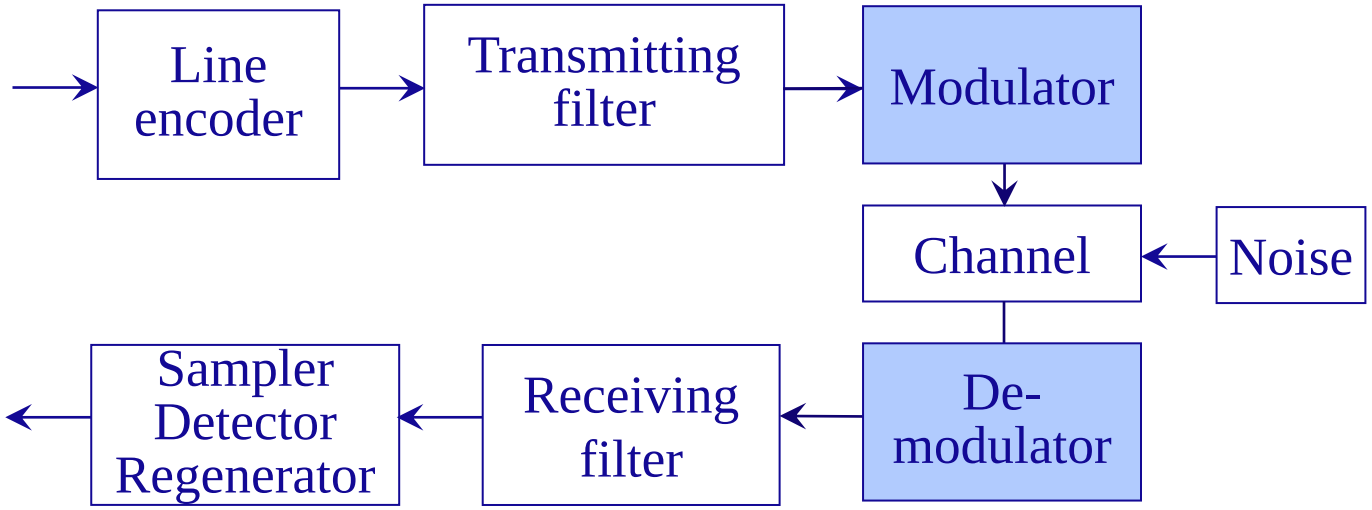
Performance comparison

Digital baseband transmission system



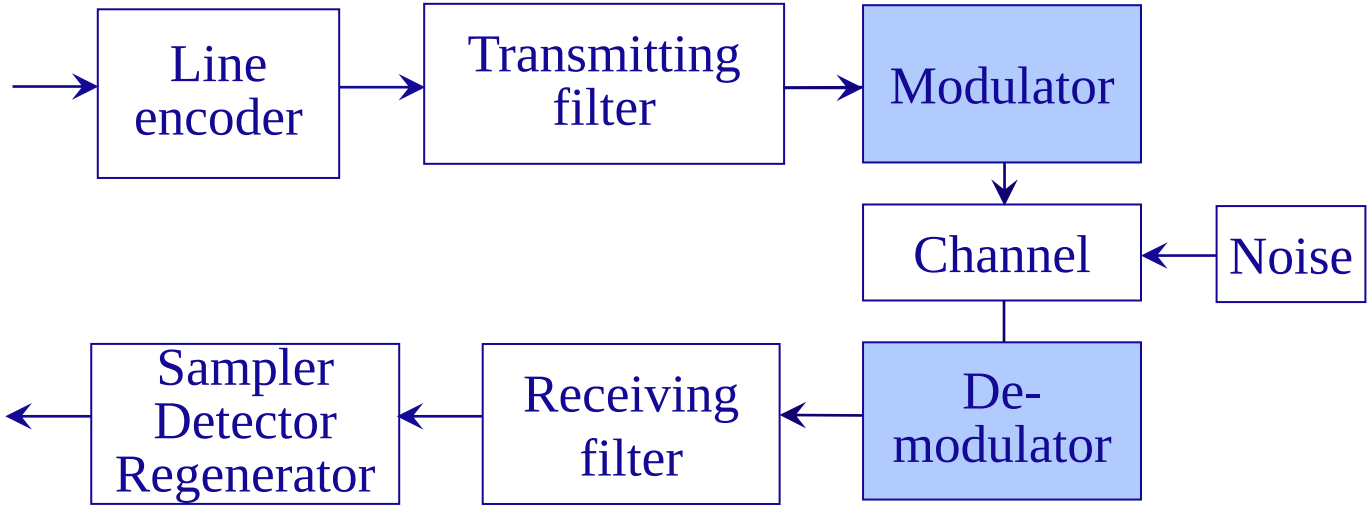
LEARNING GUIDE OF THIS CHAPTER

Digital passband transmission system



LEARNING GUIDE OF THIS CHAPTER

Digital passband transmission system



Digital Modulation

- Modulation signal: Binary or M-ary baseband signal
- Carrier: Continuous wave
- Modulator: Amplitude, frequency, phase shift keying

LEARNING GUIDE OF THIS CHAPTER

Binary Amplitude Shift Keying's

- Modulation and Demodulation
- Symbol Error Analysis

6.1 ASK

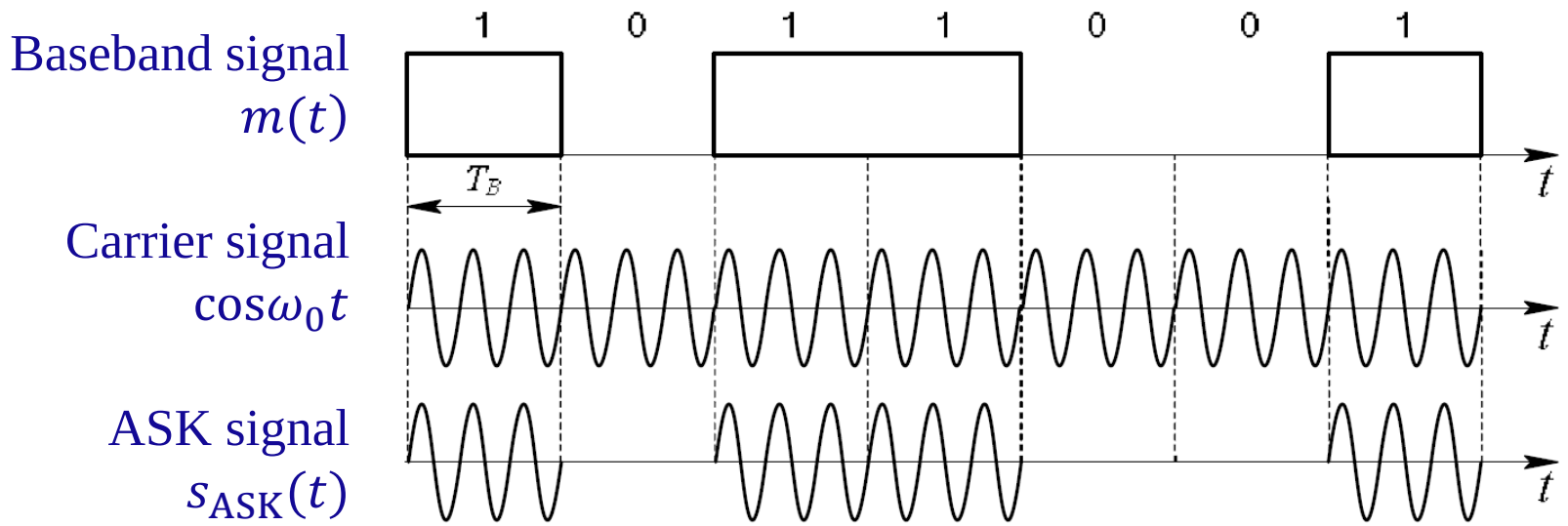
6.1.1 ASK MODULATION

ASK signal

$$s_{\text{ASK}}(t) = m(t)\cos\omega_0 t = \sum_n a_n g(t - nT_b)\cos\omega_0 t\,, \text{ where}$$

- $a_n = \begin{cases} 1 & , \text{ prob. } p \\ 0 & , \text{ prob. } 1-p \end{cases}$
- T_b – Symbol period
- $g(t)$ – Waveform
- a_n – Level value of the nth symbol

Example:



$$\begin{cases} A\cos\omega_0 t\,, & \text{if } m(t) = \text{“1”} \\ 0 & \text{, otherwise} \end{cases}$$

ASK signal's power spectrum and bandwidth

$$s_{2ASK}(t) = m(t)\cos\omega_0 t$$

- Power spectrum

$$P_{ASK}(f) = \frac{1}{4}\left[P_b(f+f_0) + P_b(f-f_0)\right]$$

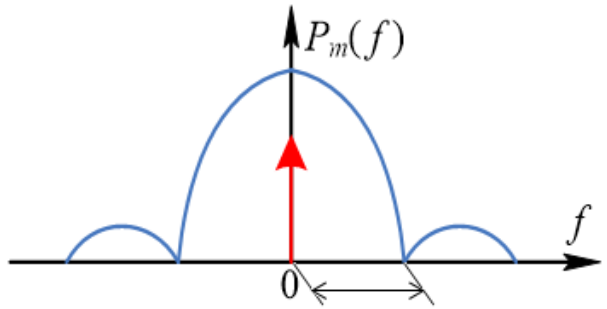
- (Effective) bandwidth

  $B_{ASK} = 2B_b = 2/T_B$ Hz

- Bandwidth efficiency

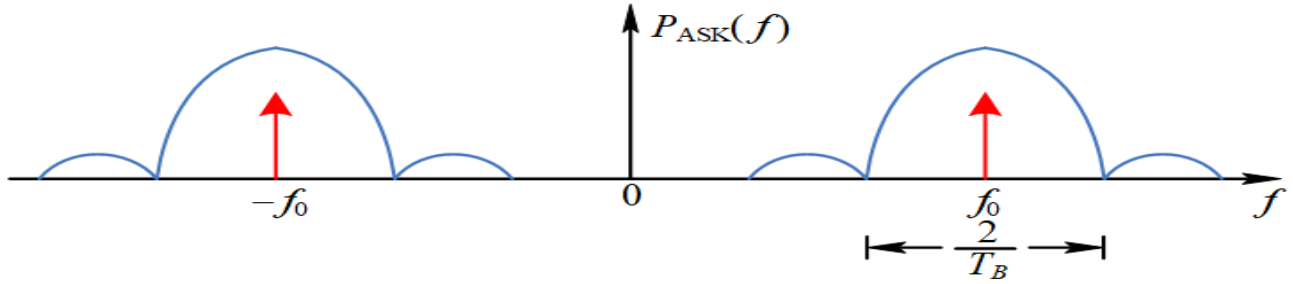
  $\eta = R_b/B = 0.5$ bit/s/Hz

- power $P_{ASK} = A^2/4$ (W)



m(t)'s bandwidth

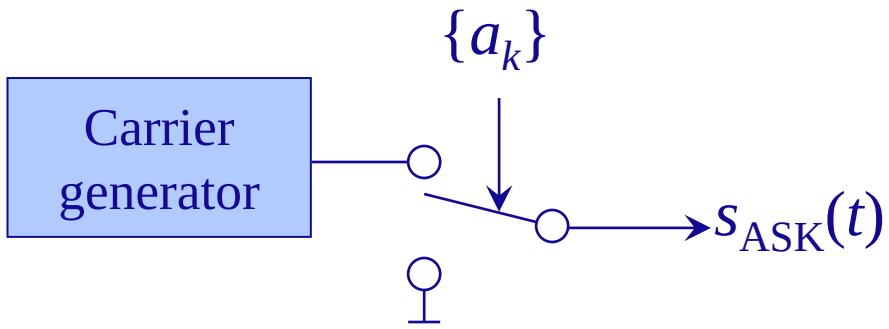
$B_b = 1/T_B = R_B$ Hz

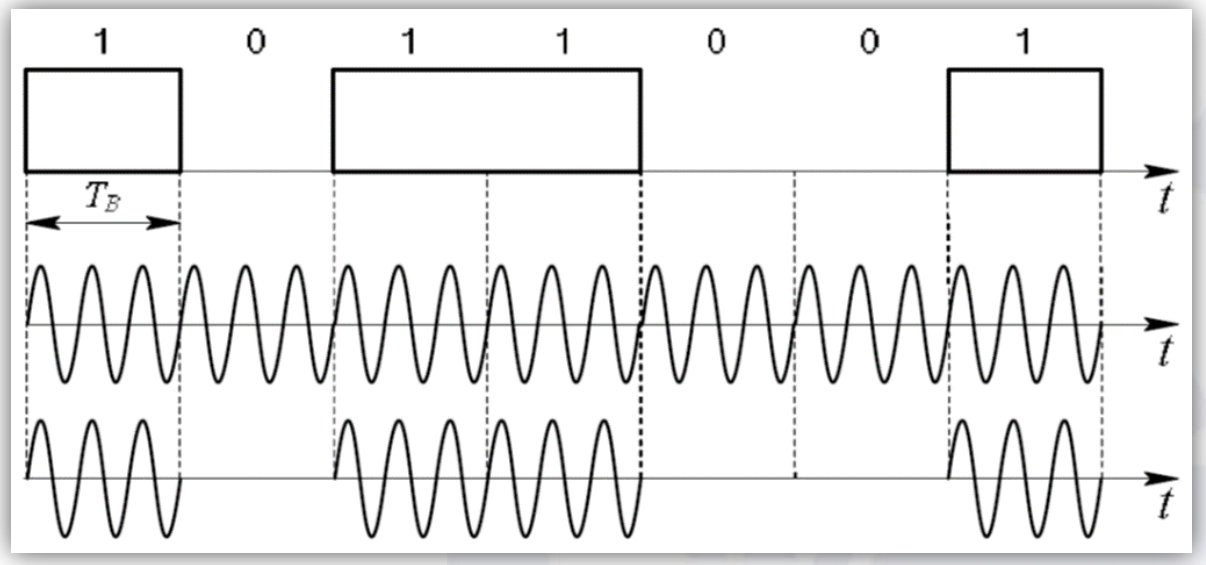


$$P_{ASK}(f) = \frac{T_B}{16}\left[Sa^2\left(\pi(f+f_0)T_B\right) + Sa^2\left(\pi(f-f_0)T_B\right)\right] + \frac{1}{16}\left[\delta(f+f_0) + \delta(f-f_0)\right]$$

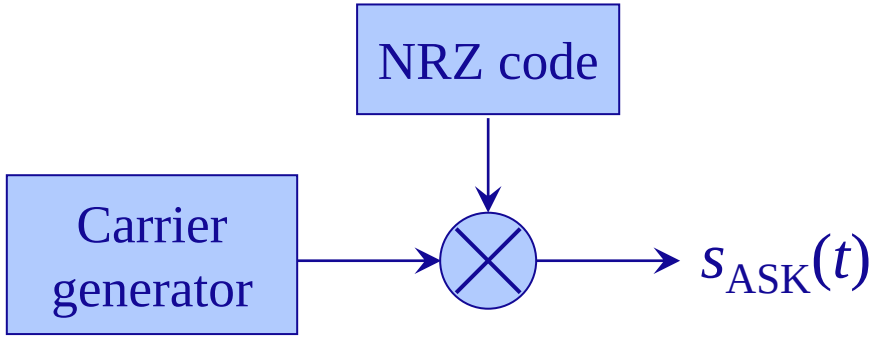
ASK modulation

- On-Off Keying (OOK)



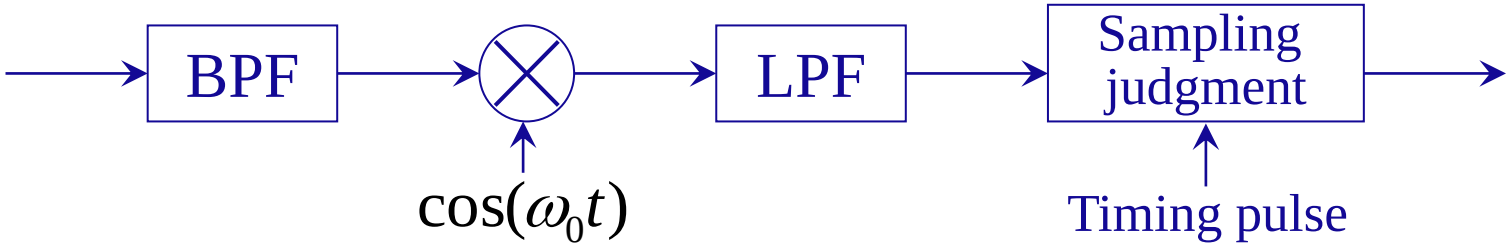


- Multiplier



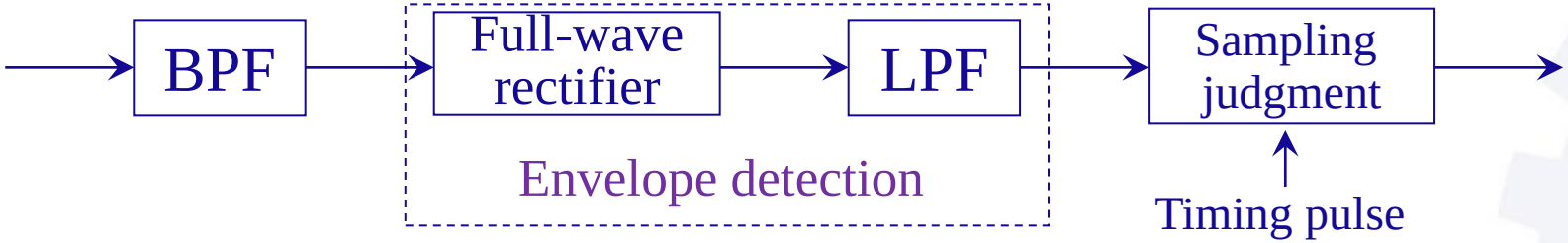
ASK demodulation

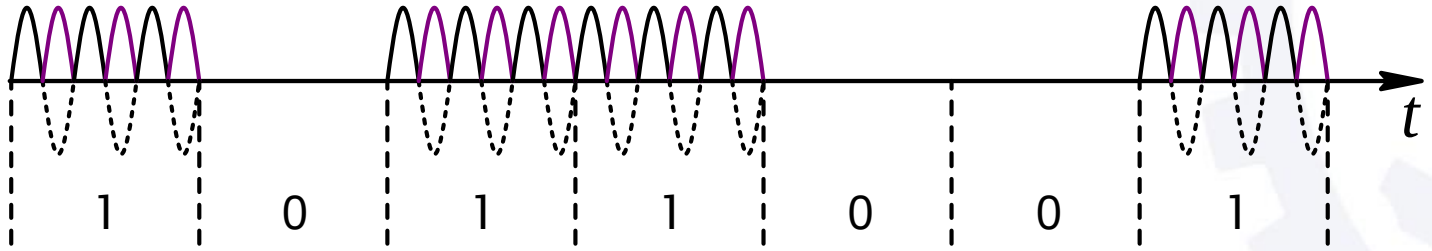
- Coherent demodulation



? Recall the coherent demodulation process of analog modulation signal

- Noncoherent demodulation





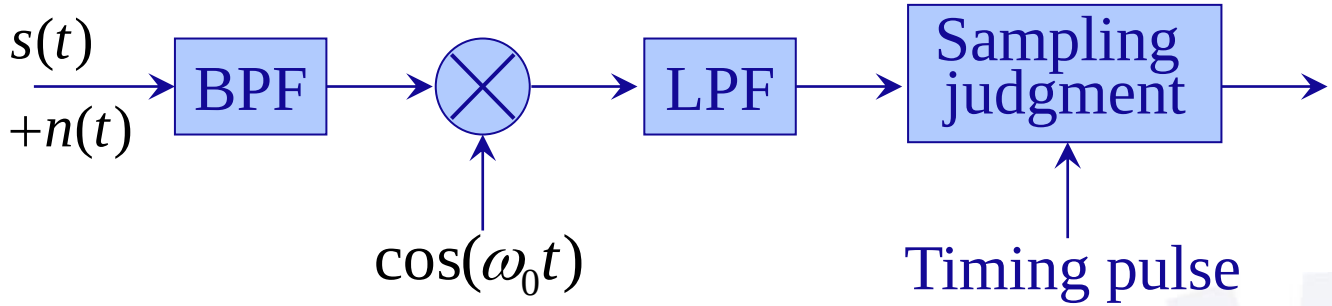
SUMMARY OF ASK MODULATION & DEMODULATION

- DSB modulation of unipolar baseband signal
- Power spectrum is the linear shift of that of the baseband signal
- Bandwidth is twice of the baseband signal's bandwidth

6.1.2 SYMBOL ERROR ANALYSIS OF ASK SIGNAL

Symbol Error Analysis of ASK Coherent Demodulation

$$s(t)=\begin{cases} A\cos\omega_0 t\,, & \text{sending "1"} \\ 0\,, & \text{sending "0"} \end{cases}$$



- BPF's output

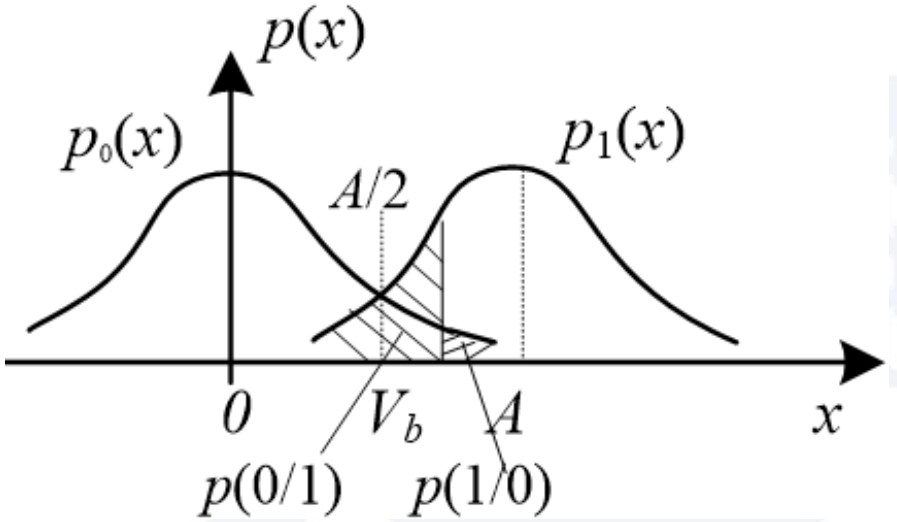
$$x(t)=s(t)+n_i(t)=\begin{cases} A\cos\omega_0 t+n_I(t)\cos\omega_0 t-n_Q(t)\sin\omega_0 t, & \text{"1"} \\ n_I(t)\cos\omega_0 t-n_Q(t)\sin\omega_0 t, & \text{"0"} \end{cases}$$

- LPF's output $x_o(t)=\begin{cases} A+n_I(t), & \text{sending "1"} \\ n_I(t), & \text{sending "0"} \end{cases}$

Here we ignore factor 1/2 for simplicity

- Sampling $x_o=\begin{cases} A+n_I, & \text{sending "1"} \\ n_I, & \text{sending "0"} \end{cases}$

Gaussian distribution



? Recall the error analysis of digital baseband transmission

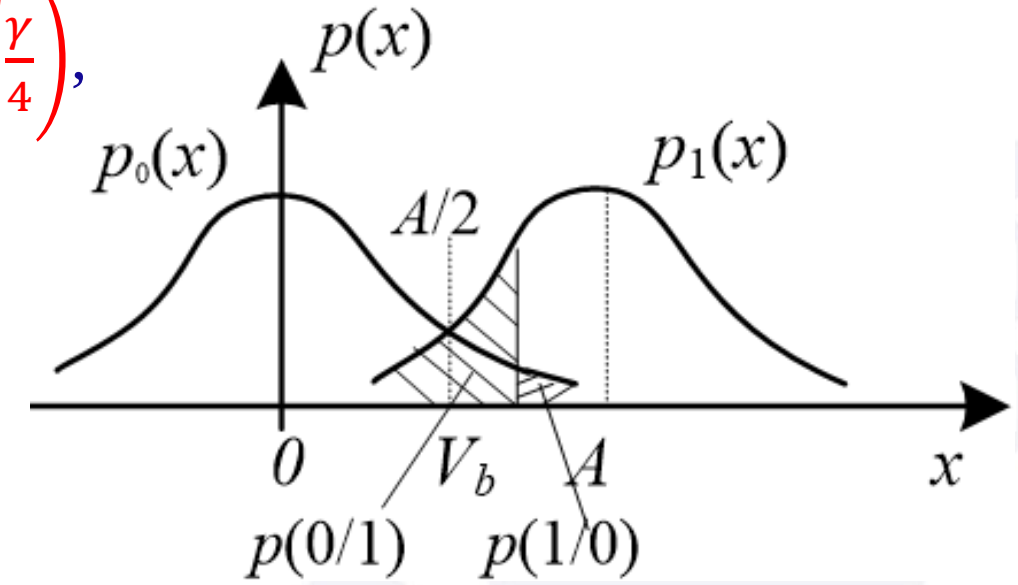
Symbol error analysis of ASK coherent demodulation

$$P_e = P(1)P(0/1) + P(0)P(1/0) = P(s_1)\int_{-\infty}^{V_b} p_1(x_1)\,\mathrm{d}x_1 + P(s_0)\int_{V_b}^{\infty} p_0(x_0)\,\mathrm{d}x_0$$

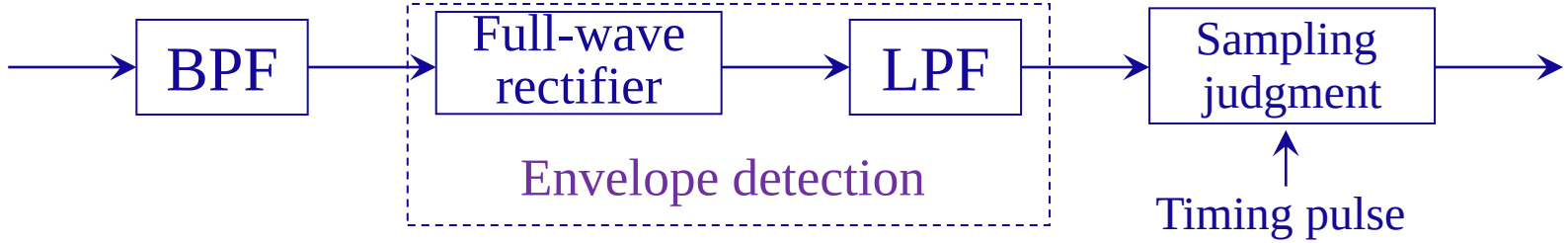
Under equal probability condition P(0)=P(1), we have

- Optimal threshold level $V_{bo} = \frac{A}{2}$,
- Symbol error rate $P_e = \frac{1}{2}erfc\left(\frac{A}{2\sqrt{2\sigma^2}}\right) = \frac{1}{2}erfc\left(\sqrt{\frac{\gamma}{4}}\right)$,

where $\sigma^2 = n_0 B_{ASK} = 2n_0 R_B$, and $\gamma = \frac{A^2}{2\sigma^2}$



Symbol error analysis of ASK noncoherent demodulation

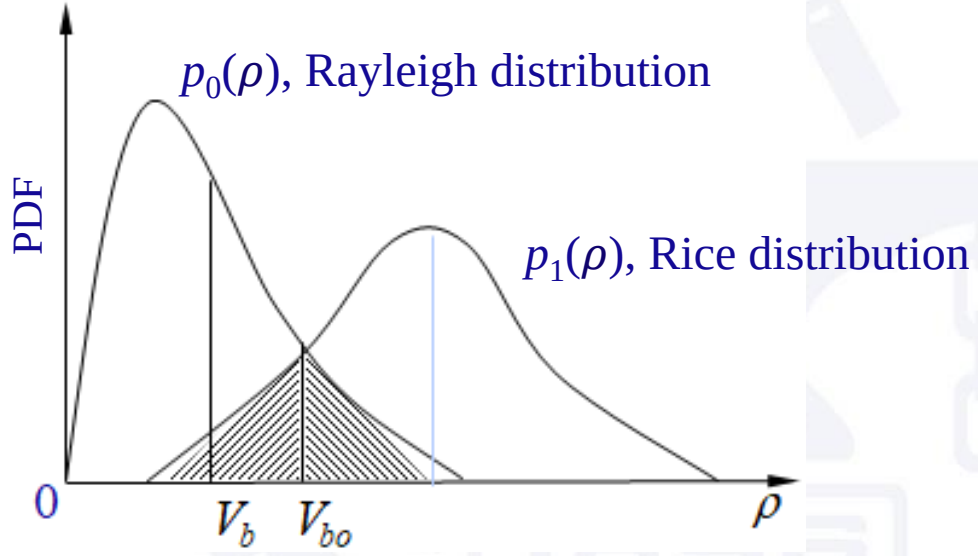


- BPF's output $x(t) = s(t) + n_i(t) = \begin{cases} A\cos\omega_0 t + n_I(t)\cos\omega_0 t - n_Q(t)\sin\omega_0 t, & \text{sending "1"} \\ n_I(t)\cos\omega_0 t - n_Q(t)\sin\omega_0 t, & \text{sending "0"} \end{cases}$

- Envelope detection's output

$$\begin{cases} \rho_1(t) = \sqrt{[A + n_I(t)]^2 + n_Q^2(t)} \\ \rho_0(t) = \sqrt{n_I^2(t) + n_Q^2(t)} \end{cases}$$

- Sampling $\rho = \begin{cases} \rho_1 & \text{, sending "1"} \\ \rho_0 & \text{, sending "0"} \end{cases}$



Symbol error analysis of ASK noncoherent demodulation

$$P_e = P(1)\int_{-\infty}^{V_b} p_1(x)\,dx + P(0)\int_{V_b}^{\infty} p_0(x)\,dx$$

$$p_0(x) = \frac{x}{\sigma_n^2}\exp(-\frac{x^2}{2\sigma_n^2})$$

$$p_1(x) = \frac{x}{\sigma_n^2}\exp(-\frac{A_0^2 + x^2}{2\sigma_n^2}) \cdot I_0(\frac{A_0 x}{\sigma_n^2})$$

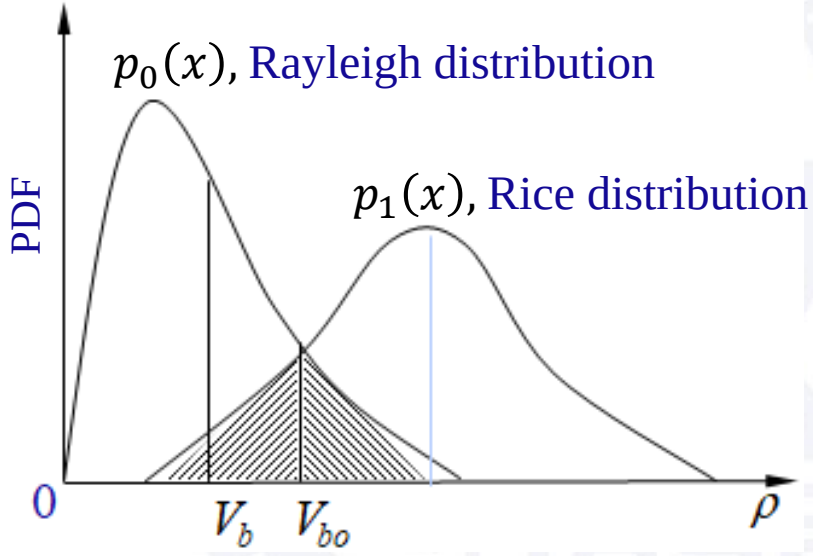
Using $P(1)p_1(V_{bo})=P(0)p_0(V_{bo})$, With large SNR x >>1, we have

$$I_0(x) \approx \frac{e^x}{\sqrt{2\pi x}} \text{ and } V_{bo} \approx \frac{A}{2}\left(1 + \frac{8\sigma^2}{A^2}\right)^{\frac{1}{2}}$$

Furthermore, under equal probability condition P(0)=P(1), we have

- Optimal threshold level $V_{bo} \approx \frac{A}{2}$,
- Symbol error rate $P_e \approx \frac{1}{2}e^{-\frac{A^2}{8\sigma^2}} = \frac{1}{2}e^{-\frac{\gamma}{4}}$,

  where where $\sigma^2 = n_0 B_{ASK} = 2n_0 R_B$ and $\gamma = \frac{A^2}{2\sigma^2}$



Comparison of coherent and noncoherent demodulation

- Optimal threshold level $V_{bo} \approx \frac{A}{2}$,

- Symbol error rates (under large SNR and equal probability condition P(0)=P(1))

$$P_{e,\text{noncoherent}} \approx \frac{1}{2} e^{-\frac{A^2}{8\sigma^2}} = \frac{1}{2} e^{-\frac{\gamma}{4}} > P_{e,\text{coherent}} = \frac{1}{2} erfc\left(\frac{A}{2\sqrt{2\sigma^2}}\right) = \frac{1}{2} erfc\left(\sqrt{\frac{\gamma}{4}}\right) \approx \frac{1}{\sqrt{\pi\gamma}} e^{-\frac{\gamma}{4}},$$

where $\sigma^2 = n_0 B_{ASK} = 2n_0 R_B$, $\gamma = \frac{A^2}{2\sigma^2}$, and $erfc(x) \approx \frac{e^{-x^2}}{\sqrt{\pi}x}$ for $x >> 1$

PERFORMANCE SUMMARY OF ASK DEMODULATION

- Symbol error rates of both coherent and noncoherent demodulations are decreasing functions of the input SNR γ
- Coherent demodulation has better performance than noncoherent demodulation
- In order to achieve the same value on P_e, the input SNR required for coherent demodulation is 3dB lower than for noncoherent demodulation

Binary Frequency Shift Keying's

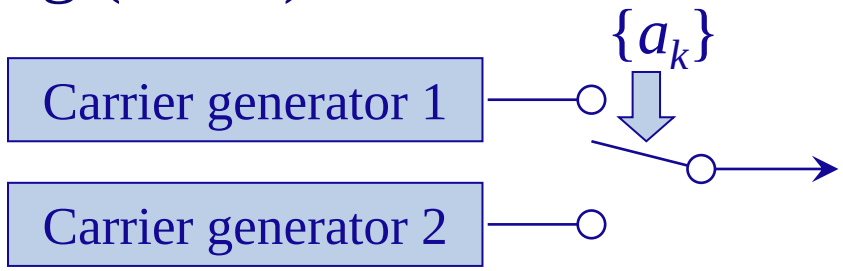
- Modulation and Demodulation
- Symbol Error Analysis

6.2 FSK

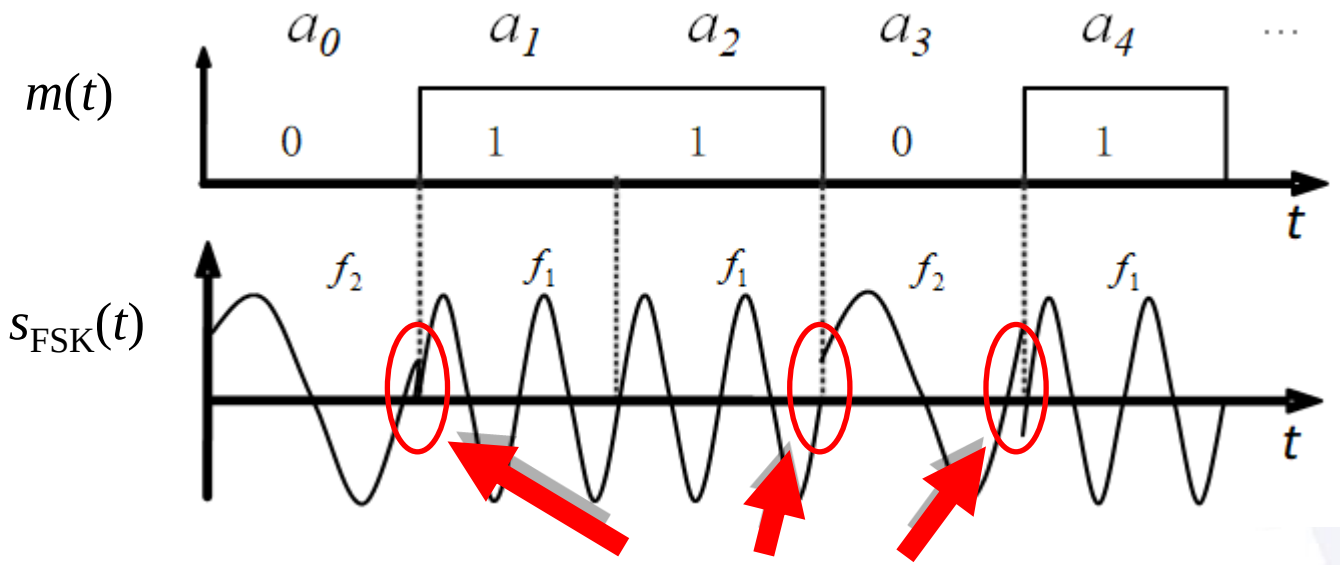
6.2.1 FSK MODULATION

FSK modulation

- On-off keying (OOK)



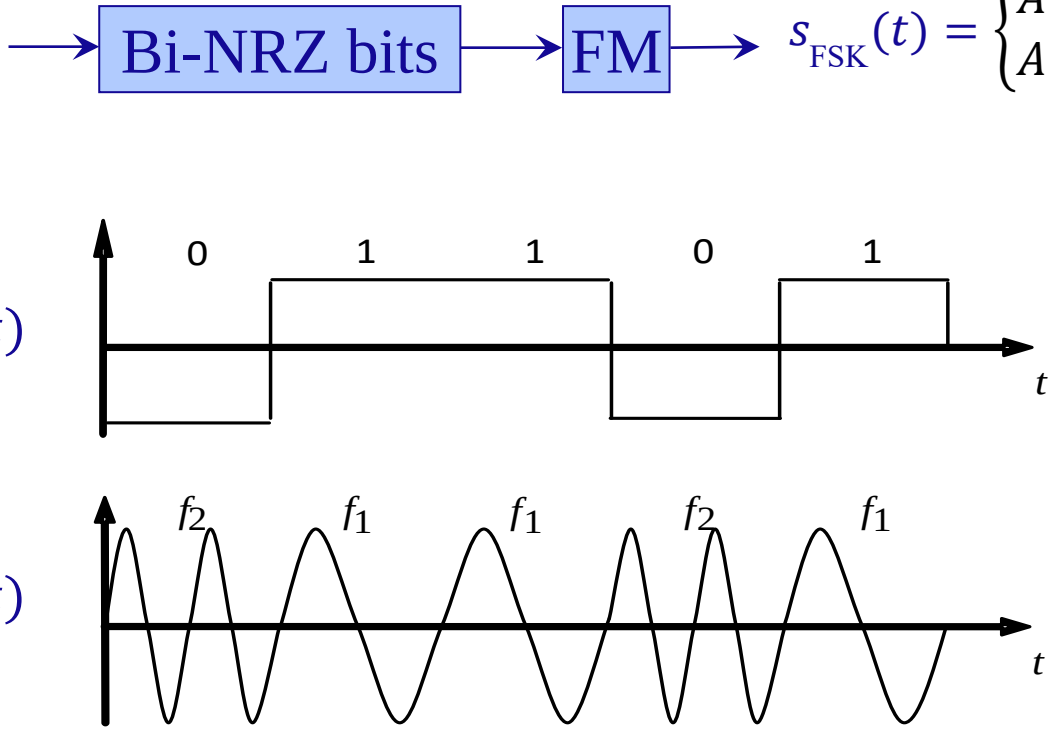
$$s_{\mathrm{FSK}}(t) = \begin{cases} A\cos(\omega_1 t + \theta_1), & \text{sending "1"} \\ A\cos(\omega_2 t + \theta_2), & \text{sending "0"} \end{cases}$$

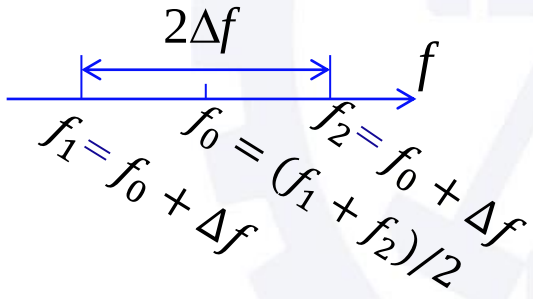


Phase discontinuity

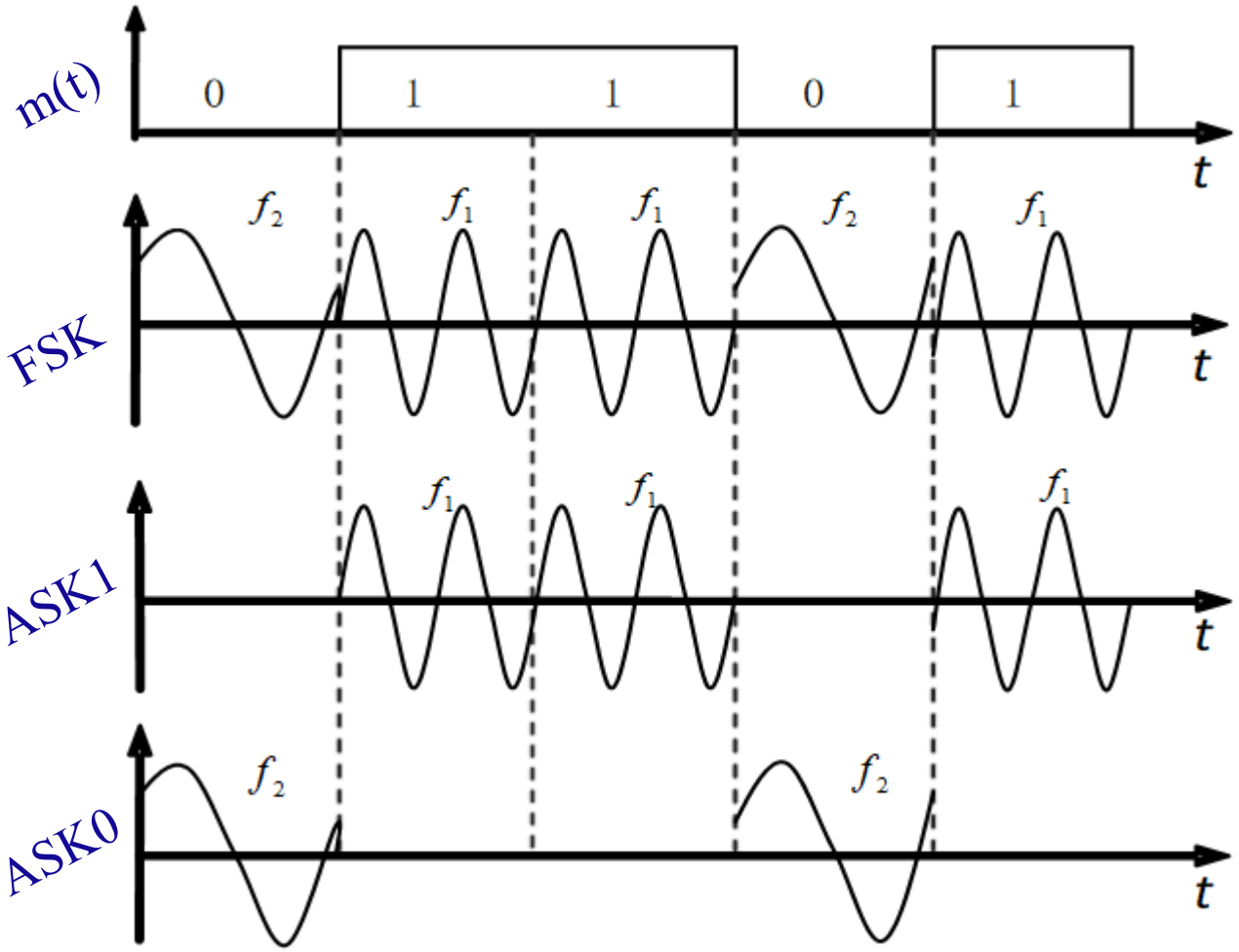
FSK modulation

- Direct frequency modulation
  - Analog modulation, Voltage Controlled Oscillator VCO
  - continuous phase (CP) FSK





Waveform of FSK signal



$$m(t) = \sum_n a_n g(t - nT_B)$$

$$s_{FSK}(t) = \begin{cases} A\cos(\omega_1 t + \theta_1), \text{for } an = 1 \\ A\cos(\omega_2 t + \theta_2), \text{for } an = 0 \end{cases}$$

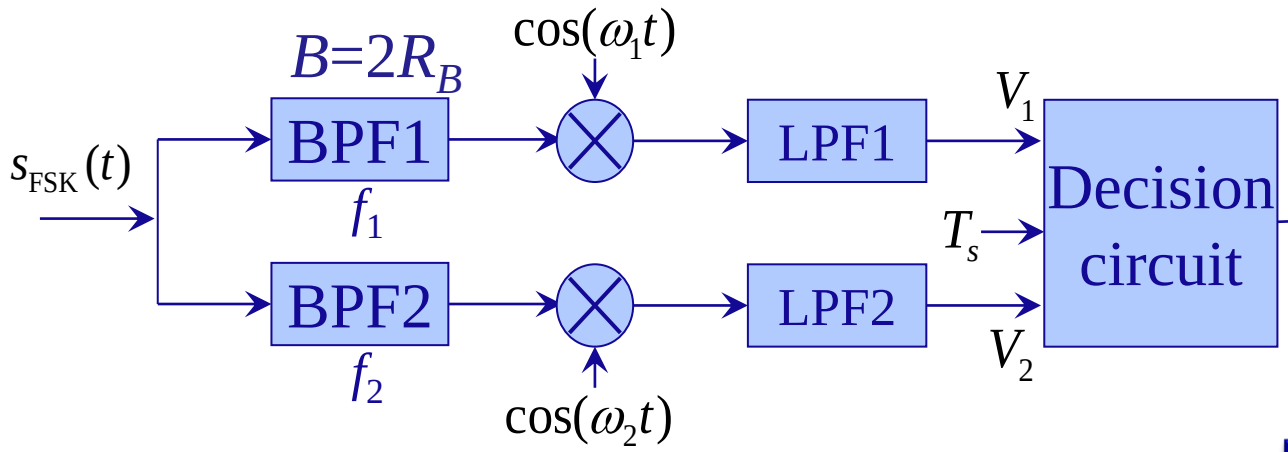
$$= \left[\sum_n a_n g(t - nT_B)\right] \cos(\omega_1 t + \theta_1)$$

$$+ \left[\sum_n \bar{a}_n g(t - nT_B)\right] \cos(\omega_2 t + \theta_2)$$

FSK demodulation

$$s_{\text{FSK}}(t)=\begin{cases}A\cos(\omega_1 t) & \text{“1”}\\ A\cos(\omega_2 t) & \text{“0”}\end{cases}$$

- Coherent demodulation

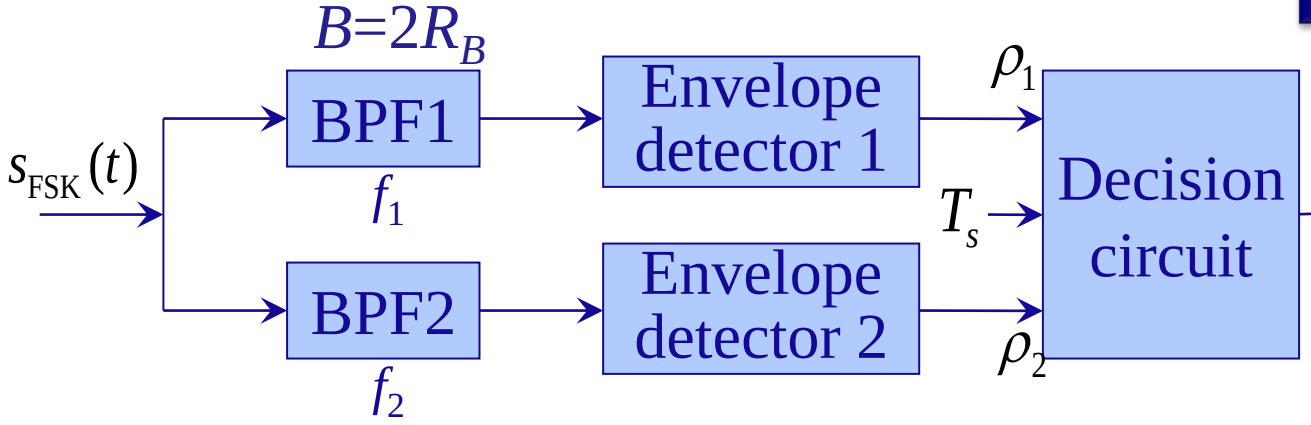


Rule:

$$\begin{cases}V_1>V_2 & \text{output "1"}\\ V_1<V_2 & \text{output "0"}\end{cases}$$

- Noncoherent demodulation

? Are there any other noncoherent demodulation methods

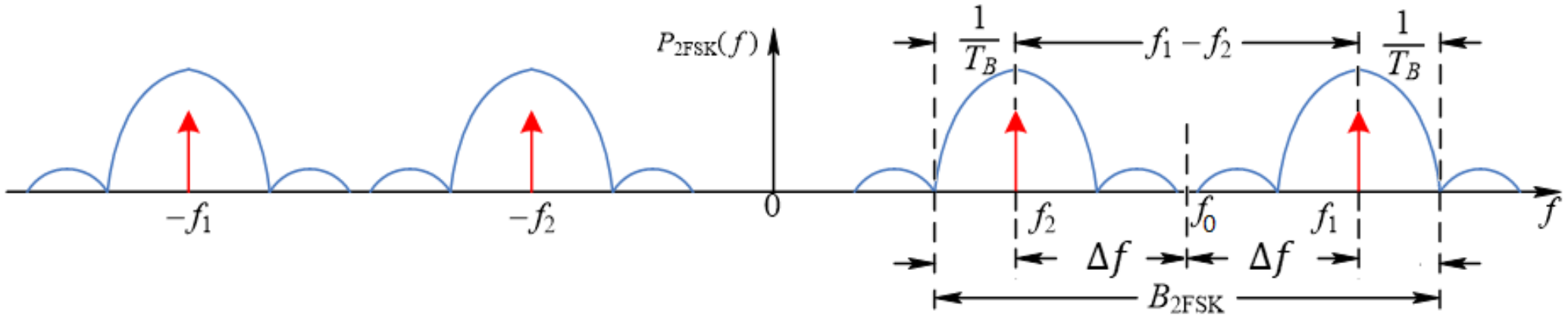


Rule:

$$\begin{cases}\rho_1>\rho_2 & \text{output "1"}\\ \rho_1<\rho_2 & \text{output "0"}\end{cases}$$

FSK signal's power spectrum

$$s_{\text{FSK}}(t) = \begin{cases} A\cos(\omega_1 t) & \text{“1”} \\ A\cos(\omega_2 t) & \text{“0”} \end{cases}$$



$$S_{\text{FSK}}(f) = \frac{T_B}{16}[Sa^2(f+f_1)T_B + Sa^2(f-f_1)T_B]$$

ASK1

$$+\frac{1}{16}[\delta(f+f_1) + \delta(f-f_1)]$$

$$+\frac{T_B}{16}[Sa^2(f+f_2)T_B + Sa^2(f-f_2)T_B]$$

ASK0

$$+\frac{1}{16}[\delta(f+f_2) + \delta(f-f_2)]$$

Effective bandwidth (According to Carson's Rule)

$$B_{2FSK} = 2R_B + |f_2 - f_1| = 2R_B + 2\Delta f = (2+h)R_B$$

For discontinuous phase $2\Delta f \geq 3R_B$

Modulation index $h = 2\Delta f / R_B$

What should be the minimal value of h ?

Average power $P_{FSK} = A^2/2$

Orthogonal design of FSK signals

$$s_{\mathrm{FSK}}(t)=\begin{cases}A\cos[2\pi(f_0+\Delta f)t] & \text{for "1"}\\ A\cos[2\pi(f_0-\Delta f)t] & \text{for "0"}\end{cases}$$

- FSK waveform

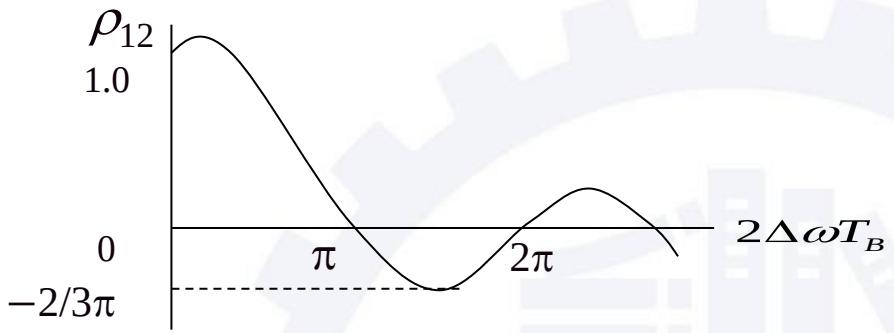
$$s_{\mathrm{FSK}}(t)=\begin{cases}A\cos[2\pi(f_0+\Delta f)t], & s_1(t)\ \text{for "1"}\\ A\cos[2\pi(f_0-\Delta f)t], & s_2(t)\ \text{for "0"}\end{cases}$$

$$E_1=\int_0^{T_B}s_1^2(t)dt;\quad E_2=\int_0^{T_B}s_2^2(t)dt;\quad E_1=E_2=\frac{A^2T_B}{2}$$

FSK has the same energy for each single "0" or "1" signal

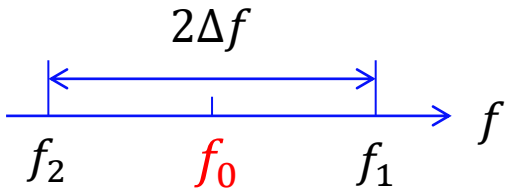
- Let correlation coefficient equal zero to make the two signals orthogonality

$$\rho_{12}=\frac{1}{\sqrt{E_1E_2}}\int_0^{T_B}s_1(t)\cdot s_2(t)dt=0,$$



we have $\begin{cases} f_0=\frac{m}{4}R_B \\ 2\Delta f=\frac{n}{2}R_B \end{cases}$,

m,*n* are positive integers



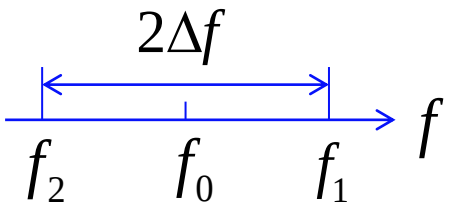
center frequency between f_1 and f_2

Orthogonal Design of FSK signals

$$s_{\mathrm{FSK}}(t)=\begin{cases}A\cos[2\pi(f_0+\Delta f)t] & \text{for "1"}\\ A\cos[2\pi(f_0-\Delta f)t] & \text{for "0"}\end{cases}$$

- Center frequency $f_0 = m\dfrac{R_B}{4}$ or $f_0 >> R_B$
- Frequency diff. $2\Delta f = n\dfrac{R_B}{2}$
- Modulation index $h=\dfrac{2\Delta f}{R_B}=\dfrac{n}{2}$

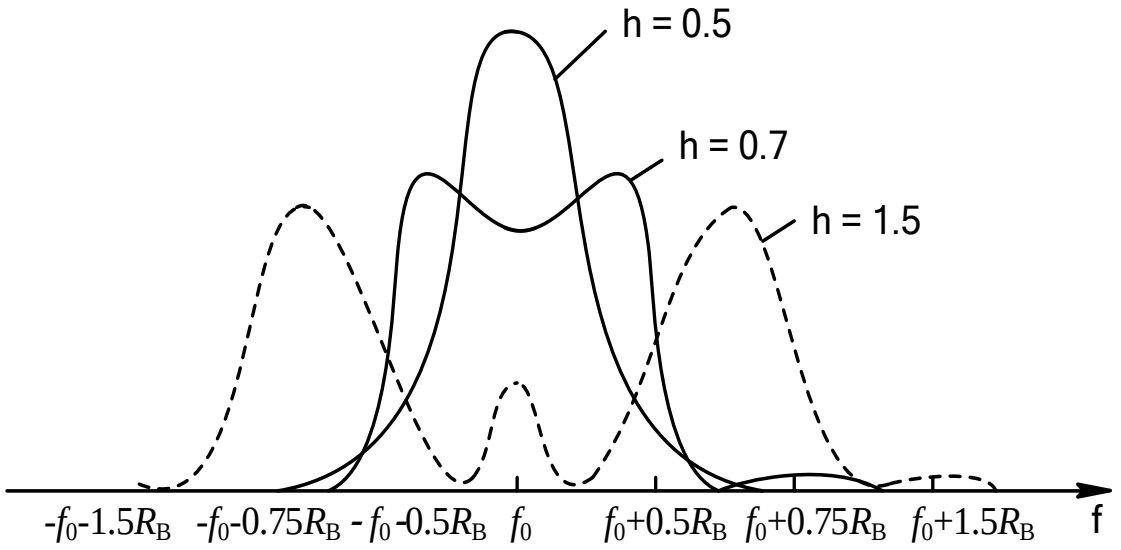
m,n positive integers



Quadrature modulation signal matches
Quadrature demodulation

When $h=\frac{1}{2}$ and $\rho_{12}=0$, we have the Minimum frequency Shift Keying (MSK), (the FSK with the minimal bandwidth)

$B_{MSK}=1.5R_B$ (Main lobe bandwidth)



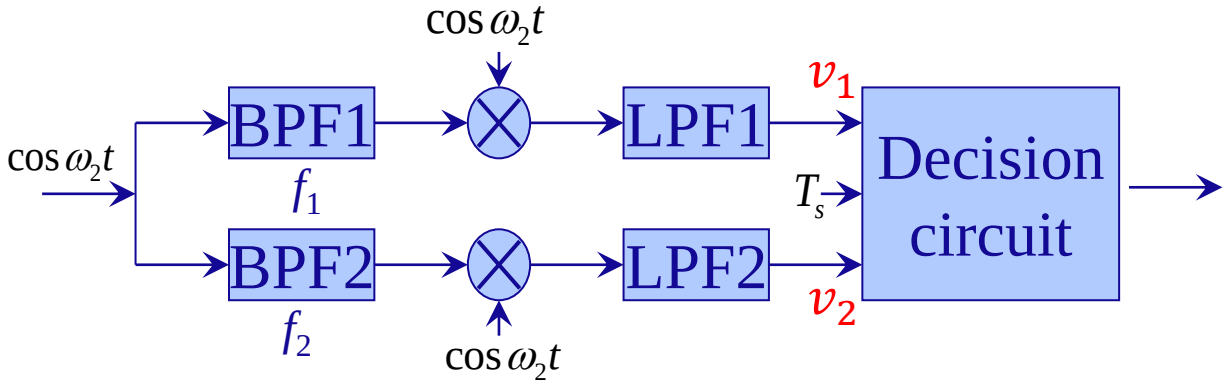
CPFSK's Power spectrum

SUMMARY OF FSK MODULATION & DEMODULATION

- FSK signal can be regarded as the sum of two ASK signals, and thus both coherent and noncoherent demodulations are applicable.
- The minimum modulation index of orthogonal of FSK signals is 1/2
- The effective bandwidth of FSK signal is determined by frequency offset and code rate
- The main lobe bandwidth of MSK signal is 1.5 times of code rate

6.2.2 SYMBOL ERROR ANALYSIS OF FSK SIGNAL

Symbol error analysis of FSK coherent demodulation



	when "1" is sent	when "0" is sent
Upper	$v_1(t) = A_0 + n_{I1}(t)$	$v_1(t) = n_{I1}(t)$
Lower	$v_2(t) = n_{I2}(t)$	$v_2(t) = A_0 + n_{I2}(t)$

Sampling judgment by $s_{\mathrm{FSK}}(t)$

BPF1's output

$$x_1(t) = \begin{cases} \left[A_0 + n_{\mathrm{I1}}(t)\right]\cos(\omega_1 t + \theta) - n_{\mathrm{Q1}}(t)\sin(\omega_1 t + \theta) & \text{, if "1" is sent} \\ n_{\mathrm{I1}}(t)\cos(\omega_1 t + \theta) - n_{\mathrm{Q1}}(t)\sin(\omega_1 t + \theta) & \text{, if "0" is sent} \end{cases}$$

BPF2's output

$$x_2(t) = \begin{cases} n_{\mathrm{I2}}(t)\cos(\omega_1 t + \theta) - n_{\mathrm{Q2}}(t)\sin(\omega_1 t + \theta) & \text{, if "1" is sent} \\ \left[A_0 + n_{\mathrm{I2}}(t)\right]\cos(\omega_1 t + \theta) - n_{\mathrm{Q2}}(t)\sin(\omega_1 t + \theta) & \text{, if "0" is sent} \end{cases}$$

Symbol error analysis of FSK noncoherent demodulation

Sampling judgment by

$$a_k' = \begin{cases} 1 & v_1 \geq v_2 \\ 0 & v_1 < v_2 \end{cases}$$

$x = A_0 + n_{I1} - n_{I2}$ is Gaussian variable ~ $N(A_0, 2\sigma_n^2)$,

since n_{I1} and n_{I2} are Gaussian variables ~ $N(0, \sigma_n^2)$

$$\bar{x} = m_x = A_0$$

$$\sigma_x^2 = E\left[(x - m_x)^2\right] = E\left[(n_{I1} - n_{I2})^2\right]$$

$$= E\left[n_{I1}^2\right] + E\left[n_{I2}^2\right] = 2\sigma_n^2$$

$$p(x) = \frac{1}{\sqrt{4\pi}\sigma_n} e^{-\frac{(x-A_0)^2}{4\sigma_n^2}}$$

Symbol error rate

$$P_e = P_{Me} + P_{Se}$$

$$= P(\text{“1”})P(v_1 < v_2 | \text{“1”}) + P(\text{“0”})P(v_1 \geq v_2 | \text{“0”})$$

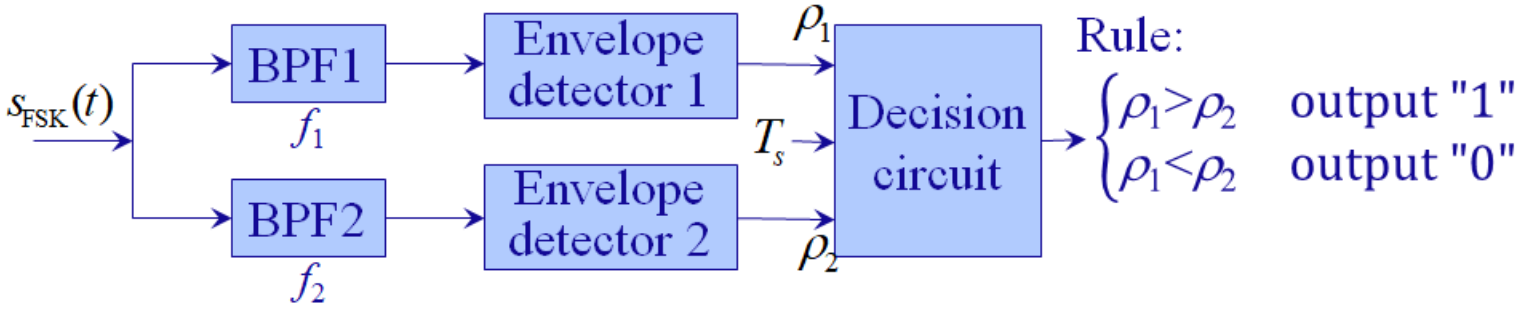
$$= P(\text{“1”})P(A_0 + n_{I1} - n_{I2} < 0) + P(\text{“0”})P(A_0 + n_{I2} - n_{I1} \leq 0)$$

$$= [P(\text{"1"}) + P(\text{"0"})] \int_{-\infty}^{0} p(x)\mathrm{d}x = \int_{-\infty}^{0} p(x)\mathrm{d}x$$

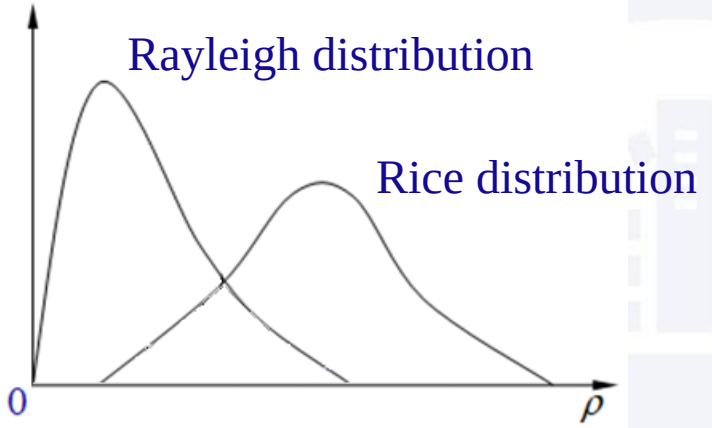
$$= \frac{1}{2}\mathrm{erfc}\left(\frac{A_0}{2\sigma_n}\right) = \frac{1}{2}\mathrm{erfc}\left(\sqrt{\frac{\gamma}{2}}\right)$$

P_e is independent of the prior probabilities $P(\text{“0”})$ and $P(\text{“1”})$

Symbol error analysis of FSK noncoherent demodulation



	when "1" is sent	when "0" is sent
Upper branch	ρ_1 ~ Rice	ρ_1 ~ Rayleigh
Lower branch	ρ_2 ~ Rayleigh	ρ_2 ~ Rice



$$P_e \approx \frac{1}{2} e^{-\frac{A^2}{4\sigma_n^2}} = \frac{1}{2} e^{-\frac{\gamma}{2}}$$

$$\gamma = \frac{A^2}{2\sigma_n^2}$$

$$\sigma_n^2 = n_0 B = 2 n_0 R_B$$

Comparison of coherent & noncoherent demodulation

- Coherent demodulation

$$P_e = \frac{1}{2} erfc\left(\frac{A}{2\sqrt{\sigma^2}}\right) = \frac{1}{2} erfc\left(\sqrt{\frac{\gamma}{2}}\right)$$

- Noncoherent demodulation

$$P_e \approx \frac{1}{2} e^{-\frac{A^2}{4\sigma^2}} = \frac{1}{2} e^{-\frac{\gamma}{2}}$$

$$\gamma = \frac{A^2}{2\sigma_n^2} \qquad \sigma_n^2 = n_0 B = 2 n_0 R_B$$

PERFORMANCE SUMMARY OF FSK DEMODULATION

- Coherent demodulation outperforms noncoherent demodulation
- No threshold is required due to comparison decision

Binary Phase Shift Keying

- Modulation and Demodulation
- Differential Phase Shift Keying (DPSK)
- Symbol Error Performance of PSK and DPSK

6.3 PSK

6.3.1 PSK MODULATION

PSK modulation methods

$$s_{\text{PSK}}(t) = A\cos(\omega_0 t + \varphi_n)$$

- A common implementation

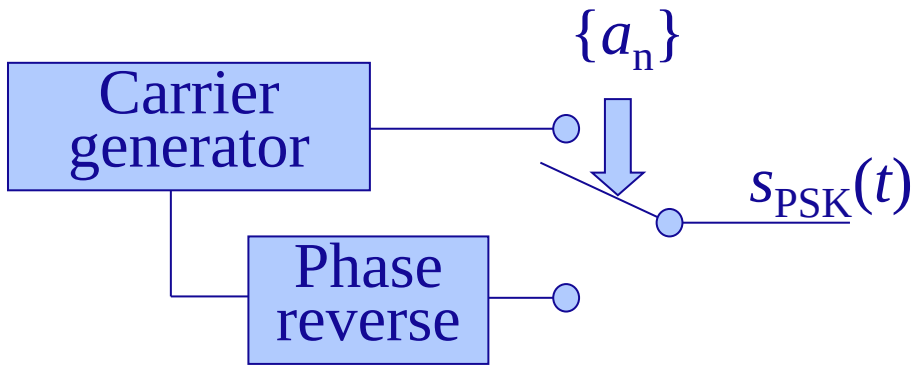
$$s_{\text{PSK}}(t) = \begin{cases} \text{A}\cos(\omega_0 t + 0) = +\text{A}\cos\omega_0 t, & \text{with prob. } P \quad \text{for "1"} \\ \text{A}\cos(\omega_0 t + \pi) = -\text{A}\cos\omega_0 t, & \text{with prob. } 1-P \text{ for "0"} \end{cases}$$

$$= \underline{\sum_n a_n g(t - nT_B)}\cos\omega_0 t, \text{ where } a_n\text{=1 or -1 for “1” or “0”}$$

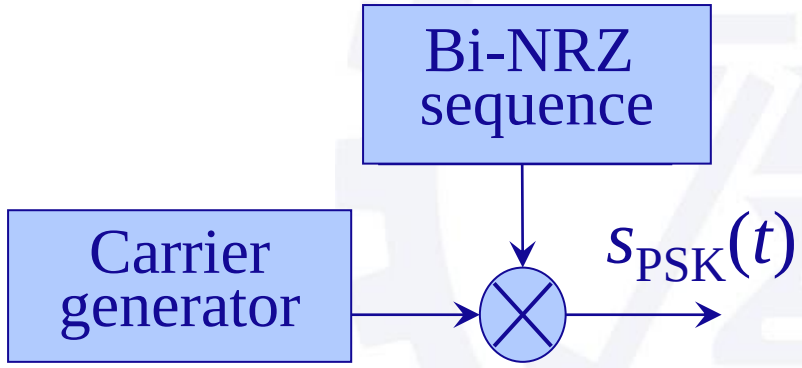
$m(t)$

- Modulation methods

1. On-off keying (Phase selection method)

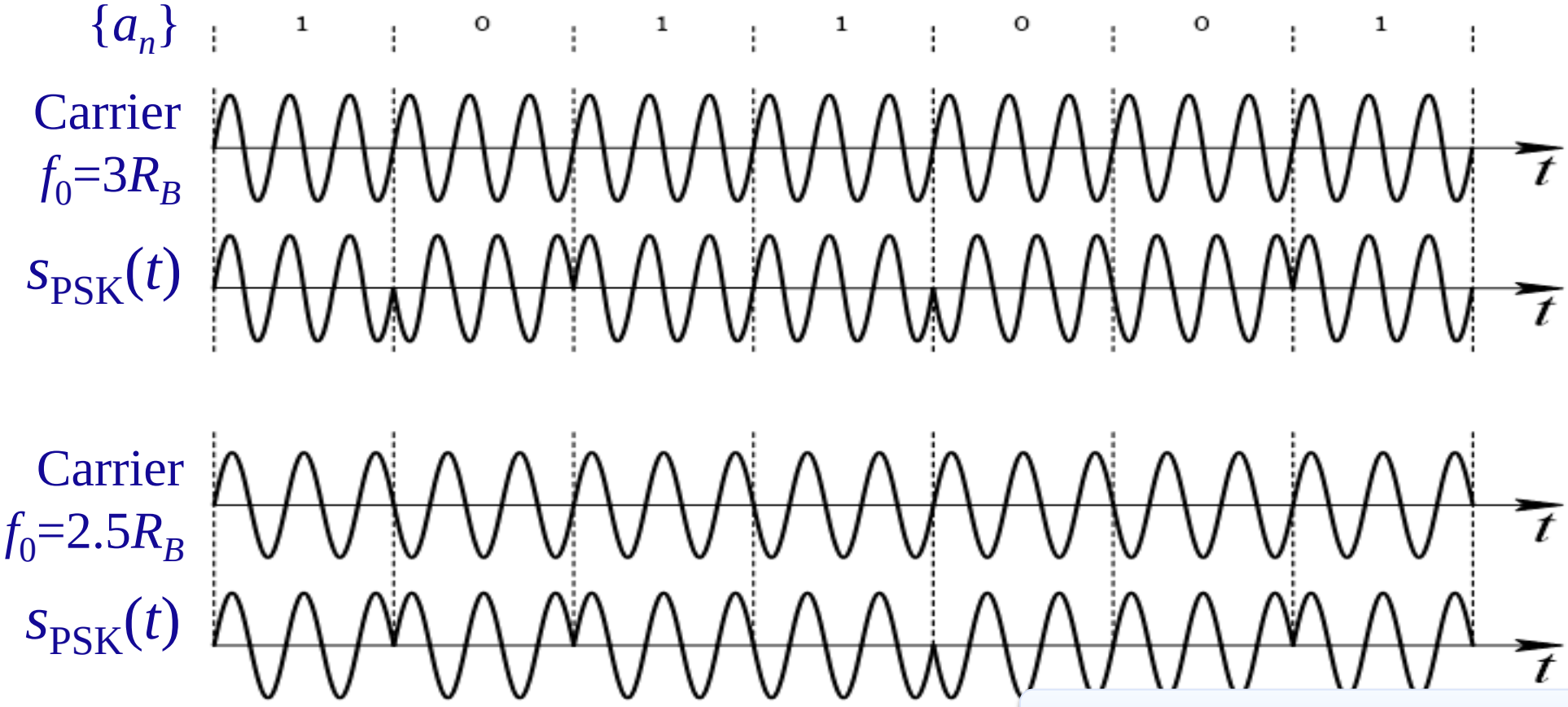


2. Multiplier (Direct phase modulation)



Waveform of PSK signal

$$s_{\text{PSK}}(t) = m(t)\cos\omega_0 t = \sum_n a_n g(t - nT_B)\cos\omega_0 t, \text{ where } a_n = \begin{cases} +1, & \text{for "1"} \\ -1, & \text{for "0"} \end{cases}$$



Phase reverse as 1/ 0 alternate

Power spectrum of PSK signal

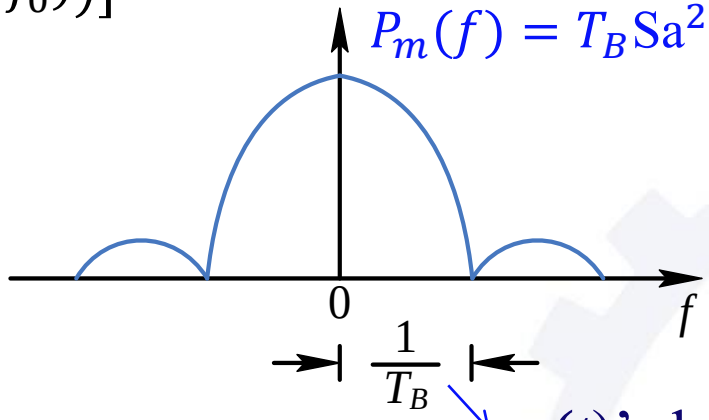
$$P_{\text{PSK}}(f) = \frac{1}{4}[P_m(f+f_0) + P_m(f-f_0)]$$

$$= \frac{A^2T}{4}\left[\text{Sa}^2\left(\pi T(f+f_0)\right) + \text{Sa}^2\left(\pi T(f-f_0)\right)\right]$$

$$s_{\text{2PSK}}(t) = m(t)\cos\omega_0 t$$

Bipolar NRZ code's power spectrum

$$P_m(f) = T_B\text{Sa}^2(\pi f T_B)$$



$m(t)$'s bandwidth (B_b)

- Effective bandwidth

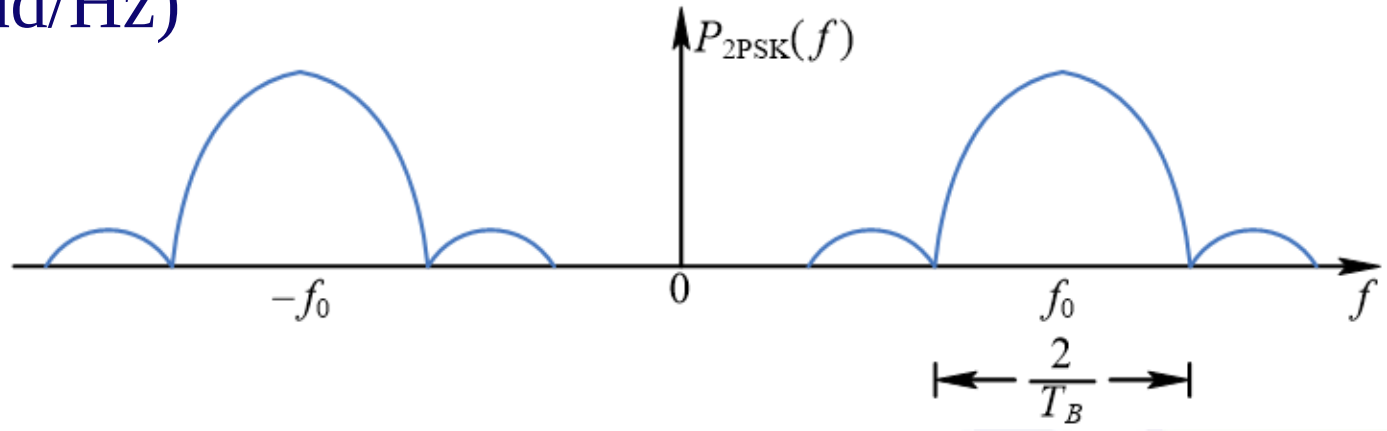
  $B_{\text{PSK}} = 2/TB = 2RB$ (Hz)

- Bandwidth efficiency

  $\eta = R_B/B_{\text{PSK}} = 1/2$ (Baud/Hz)

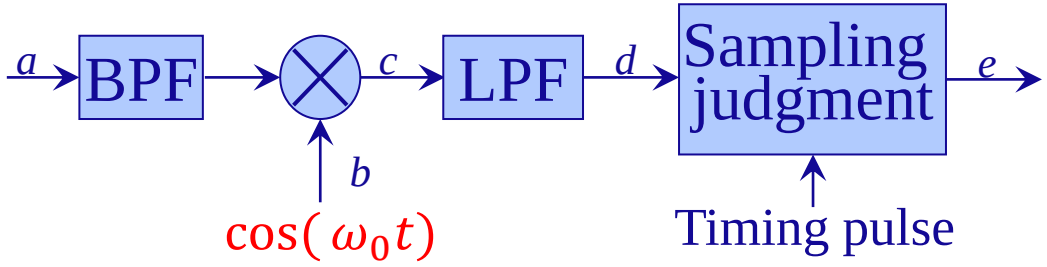
- Average power

  $P_{\text{PSK}} = A^2/2$ (W)

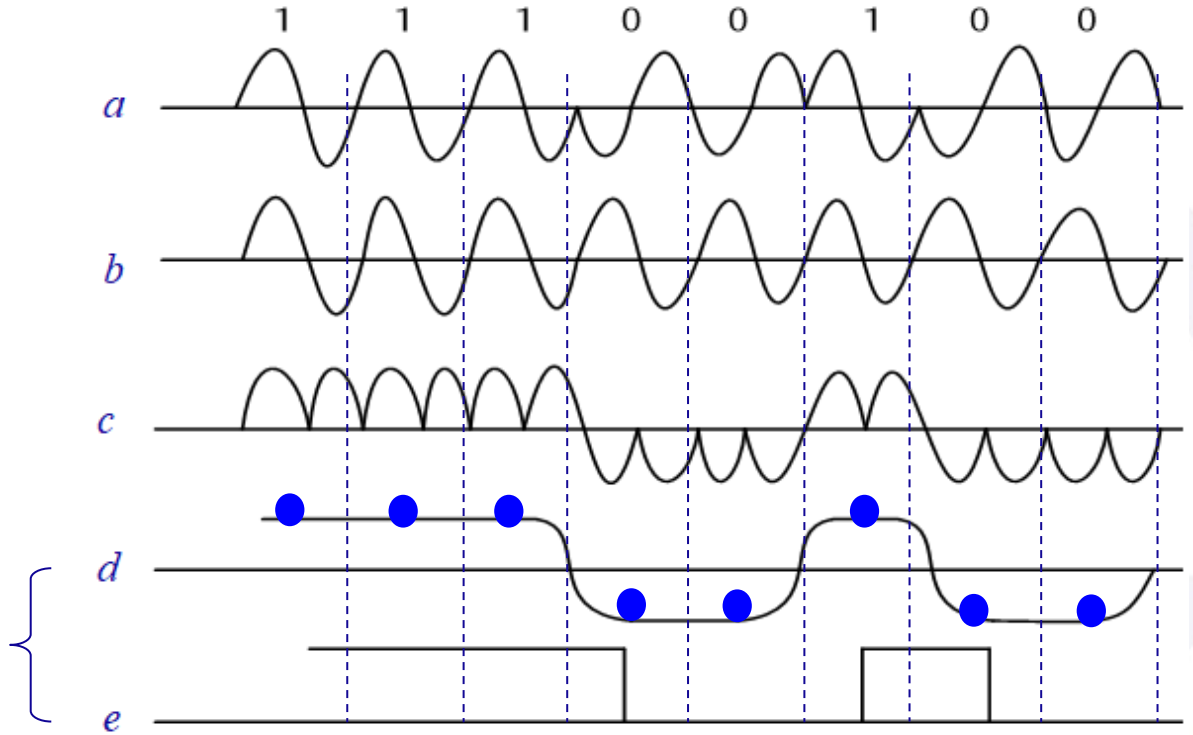


PSK demodulation methods

- Coherent demodulation



This carrier at receiver needs to be synchronized



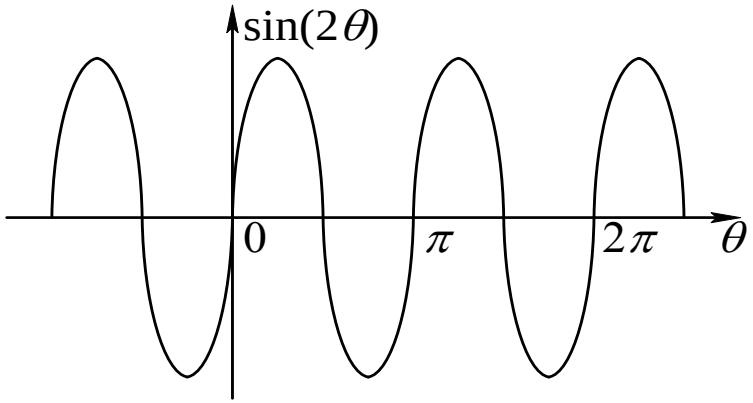
What should be decision rule ?

Synchronizing the carrier wave

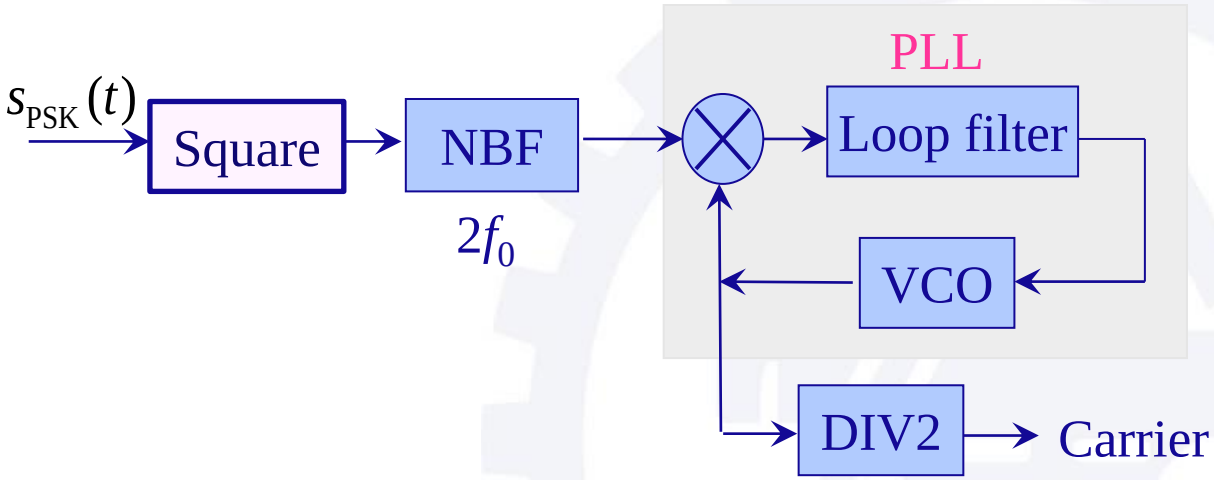
- Problem: How to obtain carrier, considering it cannot be extracted directly from PSK signal due to without discrete carrier component in its power spectrum ?

Method 1: by nonlinear transforming

$$\left[m(t)\cos(\omega_0 t)\right]^2 = \frac{1}{2}\left[m^2(t) + m^2(t)\cos(2\omega_0 t)\right]$$

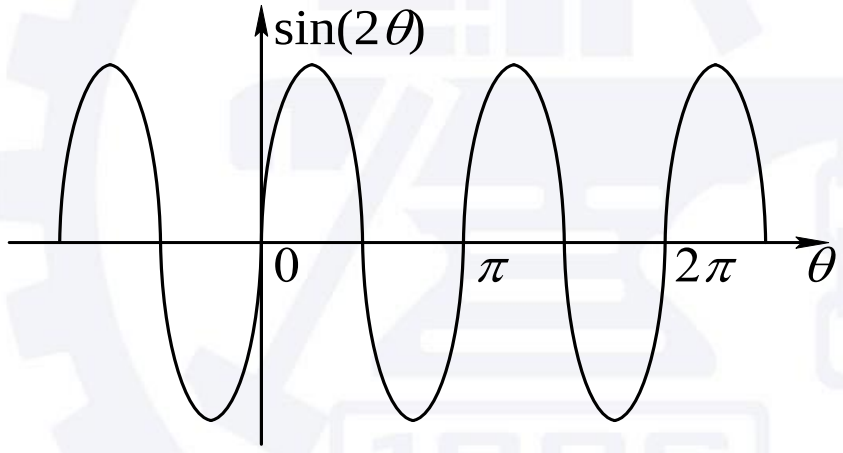


Method 2: by Square Ring, Costas Loop



Carrier Phase Ambiguity

- Problem: When using PLL to extract local reference carrier, there is phase ambiguity, which can not guarantee accurate decision
- Reason
  - PLL's phase discrimination leads to that $\theta = n\pi$ is always equilibrium point for any integer n. Thus, the phase difference between the local reference carrier and the real carrier may be 0 or π.
  - If phase difference is π, "0" and "1" will be identified completely opposite (倒π现象).



- How to overcome the influence of phase ambiguity on decision?

SUMMARY OF PSK MODULATION & DEMODULATION

- PSK is similar to ASK in modulation, while the modulation signal is bipolar code
- Only coherent demodulate is applicable, and the carrier at receiver side may have phase ambiguity (倒π现象)

6.3.2 DIFFERENTIAL PHASE SHIFT KEYING (DPSK)

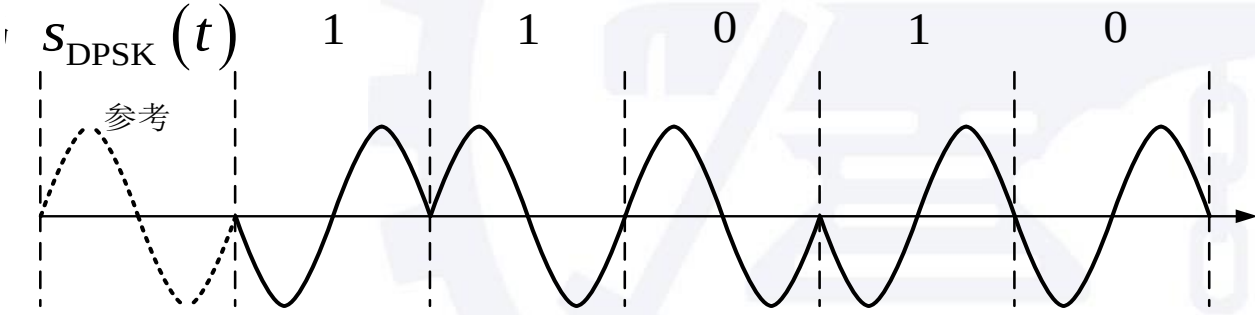
Differential Phase Shift Keying (DPSK)

Idea: Using digital baseband signal to control the carrier's phase difference $\Delta\varphi$ (between the current and previous DPSK codes), which satisfies

$$\Delta\varphi = \begin{cases} \pi, \text{corresponding to } \text{“1”} \\ 0, \text{corresponding to } \text{“0”} \end{cases}$$

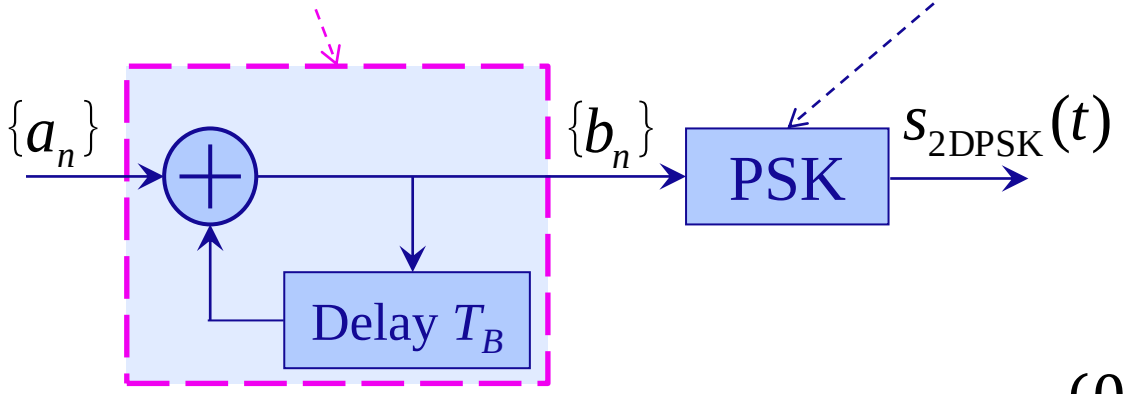
Binary sequence		1	1	0	1	0
Phase diff. $\Delta\varphi$		π	π	0	π	0
Phase (case 1)	$\varphi\ (0)$	π	0	0	π	π
Phase (case 2)	$\varphi\ (\pi)$	0	π	π	0	0

DSP signal waveform (case 1)



DPSK Modulation

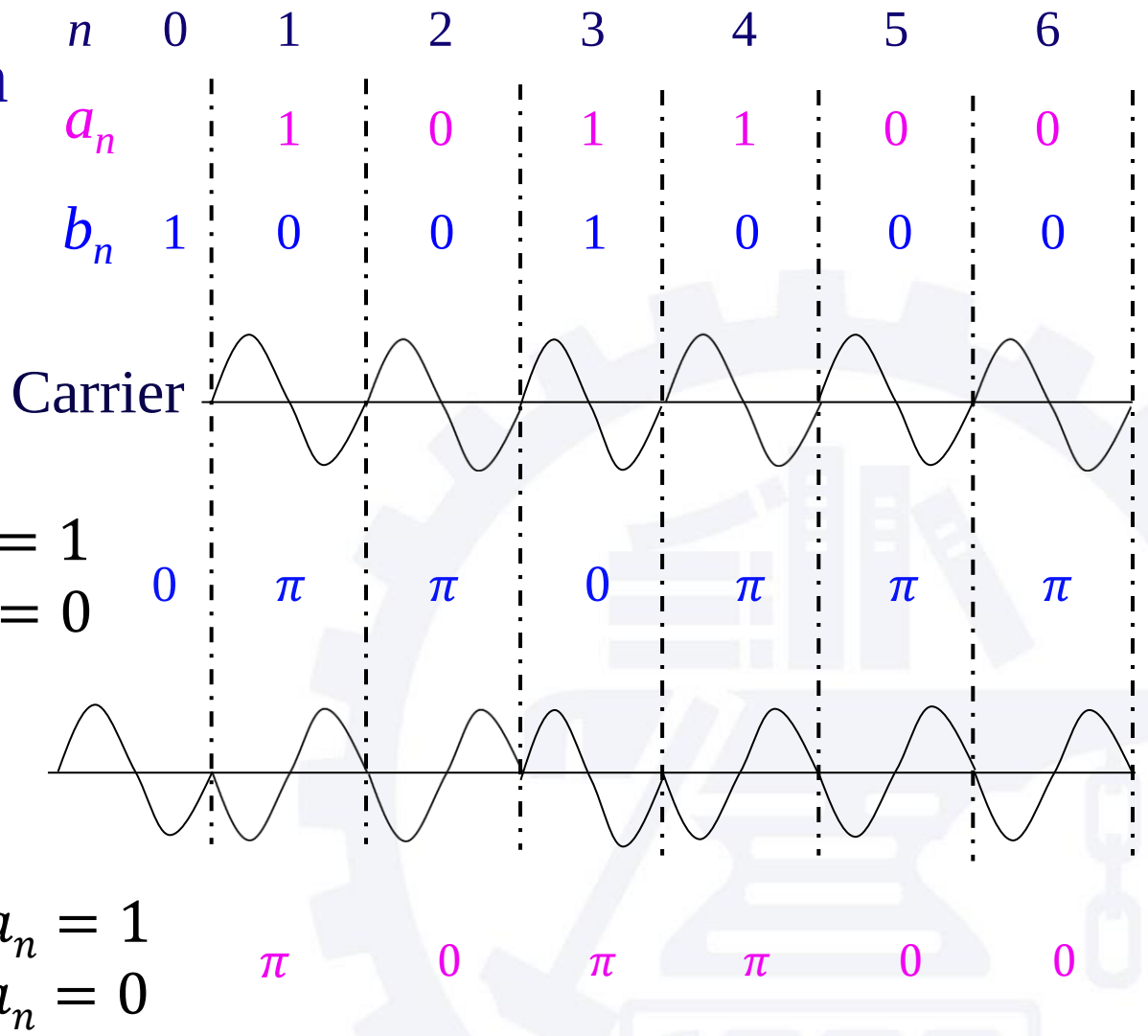
Code transformation from absolute to relative + PSK modulation



$$b_n = a_n \oplus b_{n-1}$$

"1 change/0 unchanged"

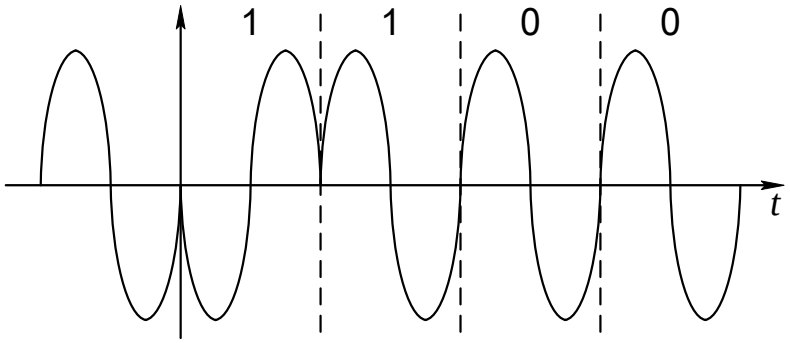
$$\text{PSK } \varphi_n = \begin{cases} 0, b_n = 1 \\ \pi, b_n = 0 \end{cases}$$



$$\text{DPSK: } \Delta\varphi = \varphi_n - \varphi_{n-1} = \begin{cases} \pi, a_n = 1 \\ 0, a_n = 0 \end{cases}$$

π 0 π π 0 0

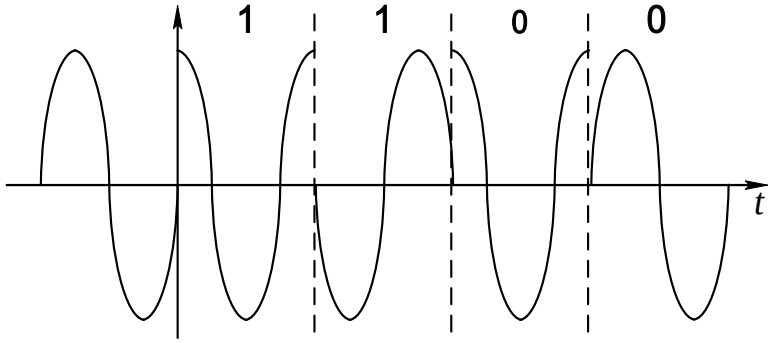
A / B mode on $\Delta\varphi$



Mode A: $\Delta\varphi = 0$ or π, represent current code is "0" or "1"



Weakness: Supposing $\{a_n\}=\{0000\ldots\}$, DPSK signal has no phase jump between adjacent symbols, and the receiver cannot correctly determine the start point in the received signal.



Mode B: $\Delta\varphi = -\pi/2$ or $-\pi/2$, represent current code is "0" or "1"

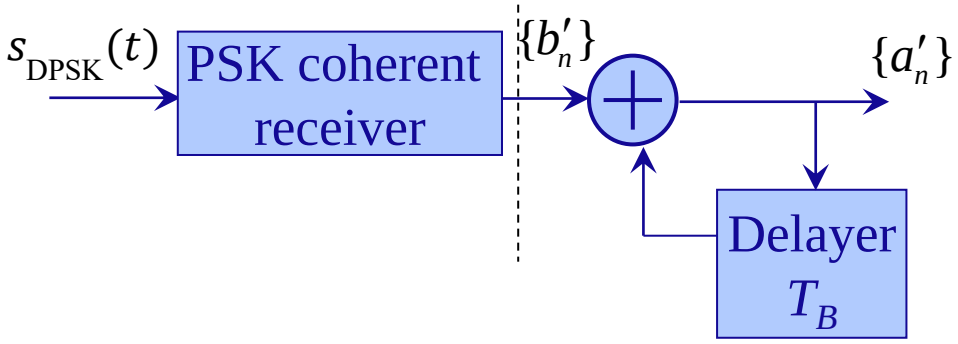


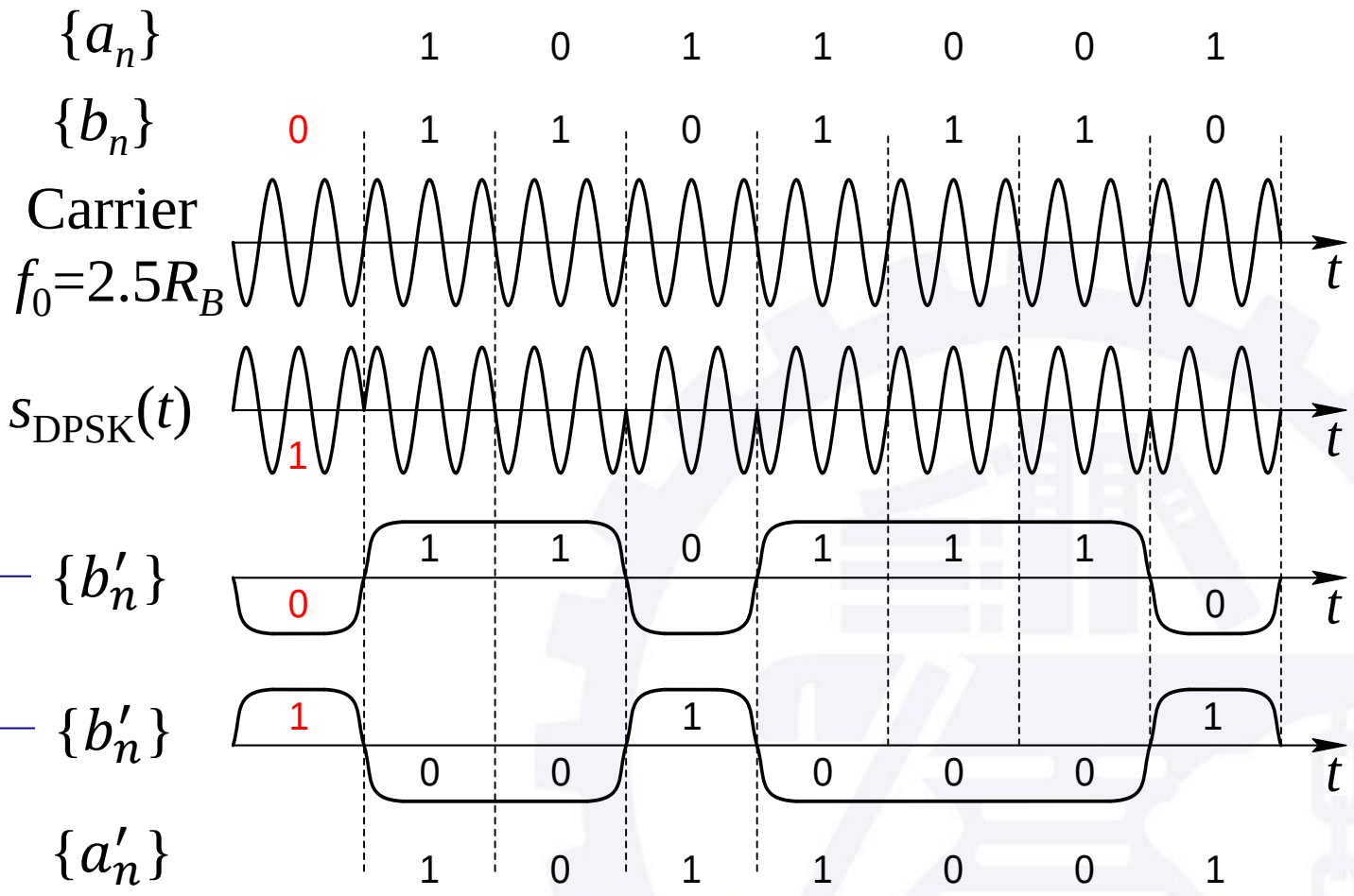
Strength: Phase must jump between any adjacent DPSK symbols, and thus the receiver can easily determine the start point

DPSK Demodulation

- Coherent demodulation

PSK demodulation + Code inverse transform $a'_n = b'_n \oplus b'_{n-1}$





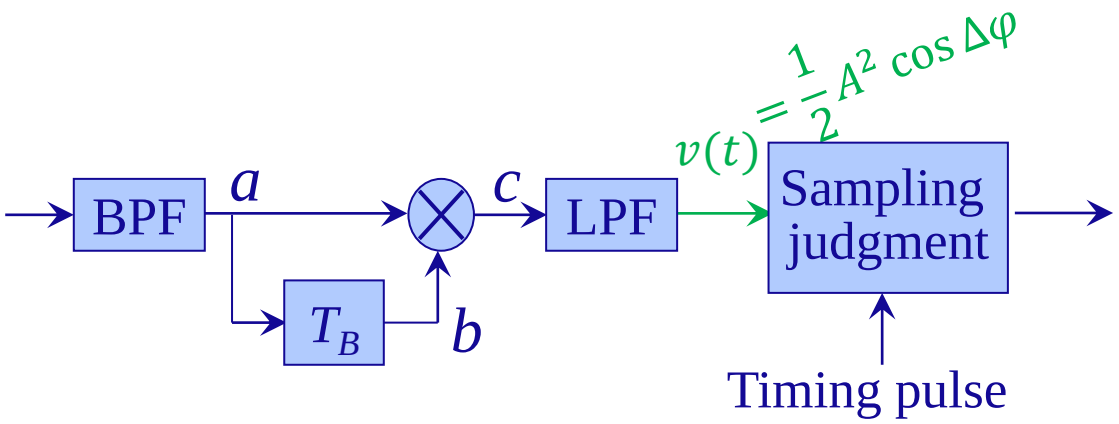
Assume it is without phase ambiguity

Assume it is with phase ambiguity

DPSK can resist phase ambiguity ? √

DPSK Demodulation

- Differential coherent demodulation (also called noncoherent demodulation)



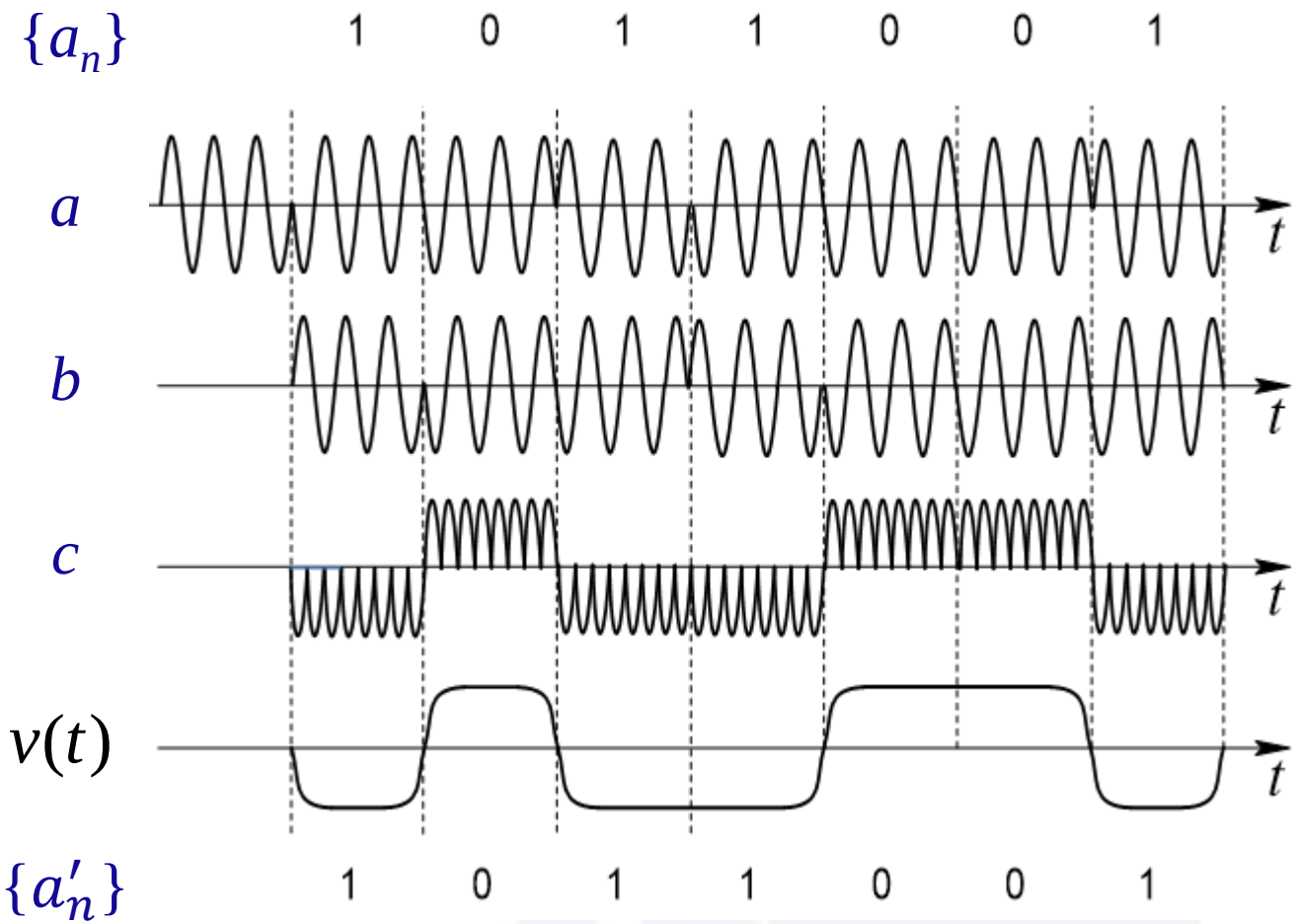
What is the decision Rule ?

$\Delta\varphi = \pi \rightarrow$ "1"

$\Delta\varphi = 0 \rightarrow$ "0"

Thus, decode $a'_n = \begin{cases} 1 & v < 0 \\ 0 & v \geq 0 \end{cases}$

Carrier frequency $f_0 = 3R_B$



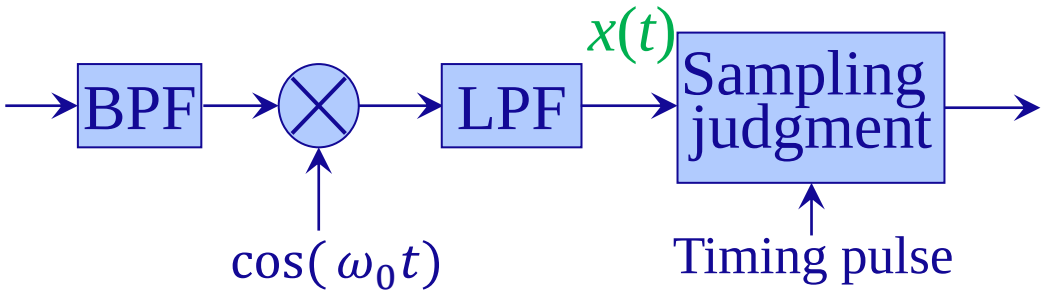
SUMMARY OF DPSK MODULATION & DEMODULATION

- DPSK carries baseband signal information by carrier phase difference
- DPSK can overcome phase ambiguity
- Both coherent demodulation and differential coherent demodulation are applicable, and the latter does not need coherent carrier

6.3.3 ERROR CODE PERFORMANCE OF PSK & DPSK

Error code performance of PSK

- Coherent demodulation

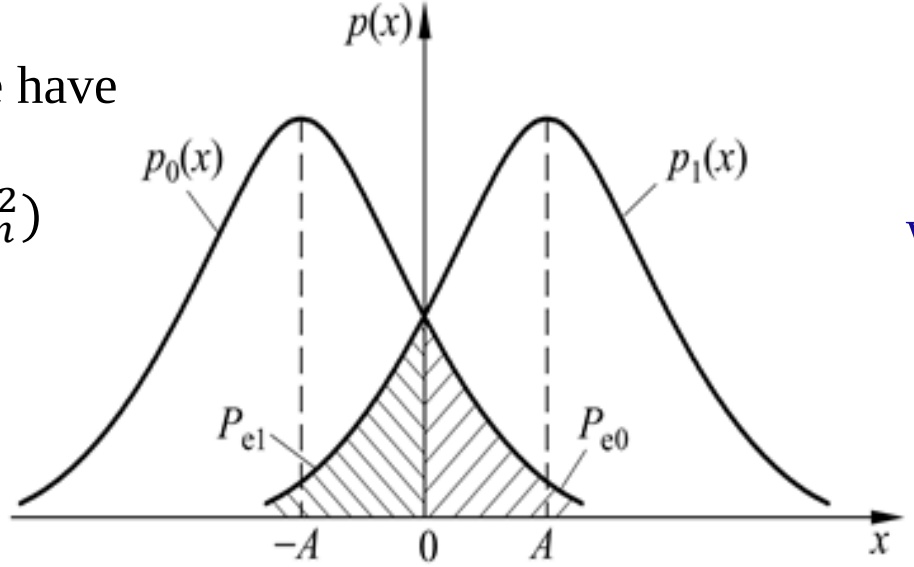


LPF's out put $x(t) = \begin{cases} A + n_I(t) & \text{if sending "1"} \\ -A + n_I(t) & \text{if sending "0"} \end{cases}$

After Sampling, we have

$p(x|1) = N(A, \sigma_n^2)$

$p(x|0) = N(-A, \sigma_n^2)$



Assuming equal prior probability $p(1)=p(0)$, the optimal threshold level is $V_{bo}=0$, independent of the level of PSK signal, and we have the minimal error code rate,

$$P_e = \frac{1}{2} erfc\left(\frac{A}{\sqrt{2}\sigma_n}\right) = \frac{1}{2} erfc(\sqrt{\gamma}),$$

where $\gamma = \frac{A^2}{2\sigma_n^2}$, and $\sigma_n^2 = n_0 B = 2n_0 R_B$

Error code performance of DPSK

- We know DPSK coherent demodulation: PSK demodulation + Code inverse transform $a'_n = b'_n \oplus b'_{n-1}$

① DPSK demodulation will work correctly only for the two cases: the current symbol and the previous symbol are both correct or both wrong, and thus has error code rate,

$$P_e = 1 - P_c = 1 - [P'P' + (1 - P')(1 - P')] = 2P' - 2P'^2,$$

where P' is PSK coherent demodulation error code rate, $P' = \frac{1}{2}\mathrm{erfc}\left(\frac{A_0}{\sqrt{2}\sigma_n}\right)$.

② Since P' is very small in general, we usually use error code rate $2P'$ for DPSK,

$$P_e = \frac{1}{2}erfc\left(\frac{A}{\sqrt{2}\sigma_n}\right) = erfc(\sqrt{\gamma}),$$

- Besides, the error code rate of differential coherent demodulation is

$$P_e = \frac{1}{2}e^{-\frac{A^2}{2\sigma_n^2}} = \frac{1}{2}e^{-\gamma}$$

(ignoring the analysis process)

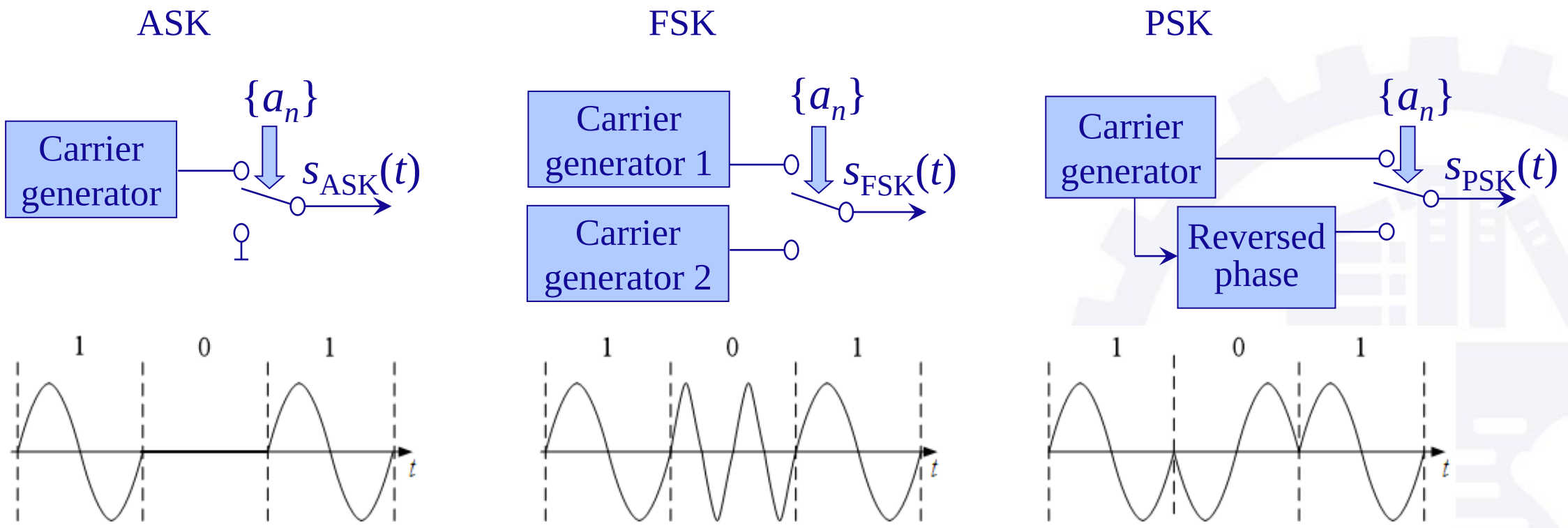
PERFORMANCE SUMMARY OF PSK/DPSK DEMODULATION

- Under equal prior probability assumption, the best threshold of PSK coherent demodulation is zero, not affected by PSK signal level
- The error rate of DPSK coherent demodulation is twice that of PSK's

6.4 PERFORMANCE COMPARISON OF BINARY DIGITAL MODULATION

On-off keying

- In on-off keying (OOK), you have a pulse shape to gate on and off to differentiate between two possible symbols



Efficiency and reliability

	Symbol Error Rate		Effective BW	Power
	Coherent	Noncoherent	Hz	W
2ASK	$\frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{\gamma}{4}}\right)$	$\frac{1}{2}e^{-\frac{\gamma}{4}}$	$2R_B$	$A^2/4$
2FSK	$\frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{\gamma}{2}}\right)$	$\frac{1}{2}e^{-\frac{\gamma}{2}}$	$2\Delta f+ 2R_B$	$A^2/2$
2PSK	$\frac{1}{2}\operatorname{erfc}(\sqrt{\gamma})$	$\frac{1}{2}e^{-\gamma}$	$2R_B$	$A^2/2$
2DPSK	$\operatorname{erfc}(\sqrt{\gamma})$	$\frac{1}{2}e^{-\gamma}$	$2R_B$	$A^2/2$

Note 1: At high SNR, coherent demodulation outperforms incoherent demodulation

Note 2: Fixed BER and demodulation method, the required γ is reduced by 3dB in turn for ASK, FSK and PSK

Note 2: Approximate formula $\operatorname{erfc}(x) \approx e^{-x^2}/(\sqrt{\pi}x)$, for $x \gg 1$

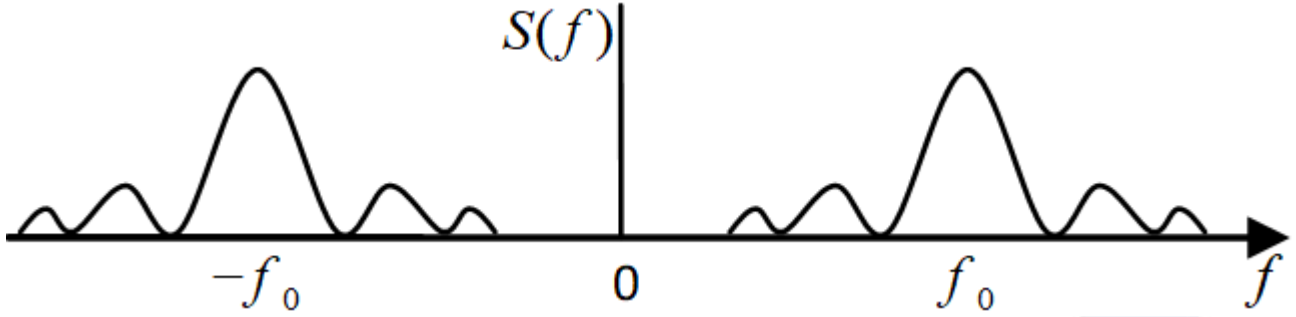
Sensitivity of demodulator to signal and noise

- Sensitivity comparison
  1. The optimal decision threshold for ASK demodulator varies with the power of received signal, so it is inconvenient.
  2. No threshold is needed by FSK demodulator, which makes decision by comparing sampling values from the upper & lower branches.
  3. Zero-value threshold is set in PSK demodulator, independent of signal power
  4. DPSK differential coherent demodulation does not need coherent carrier, and thus it is suitable for unstable channel system
- Widely used Binary digital bandpass transmission technologies at present
  1. B-DPSK with coherent demodulation: for high speed data transmission
  2. B-FSK with noncoherent demodulation, for medium and low speed data transmission

To further improve effectiveness

- To improve bandwidth utilization

1. Carrier suppressed double sideband (DSB-SC) digital amplitude modulation, realized by using Bipolar Nonreturn-to-zero Codes (BNRZ) to multiply the carrier

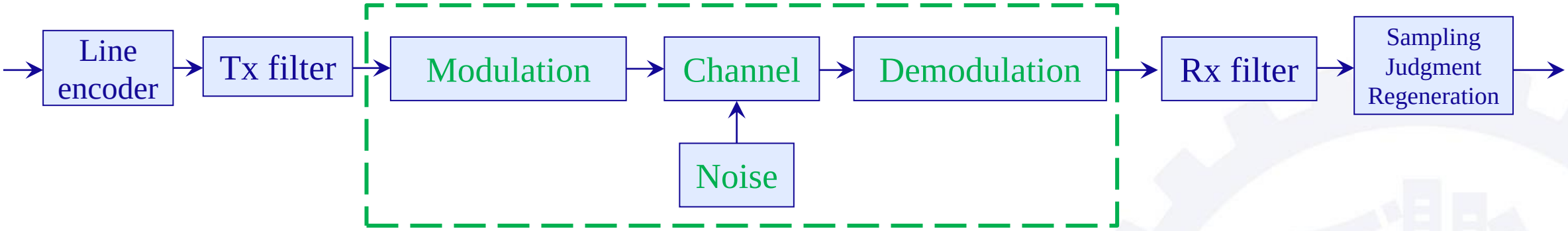


- To improve bandwidth utilization

2. Two DSB-SC digital amplitude modulated signals can form an orthogonal modulation system if their spectrums are orthogonal.
3. Single-sideband (SSB) or vestigial sideband (VSB) digital amplitude modulation

Digital bandpass transmission without intersymbol interference (ISI)

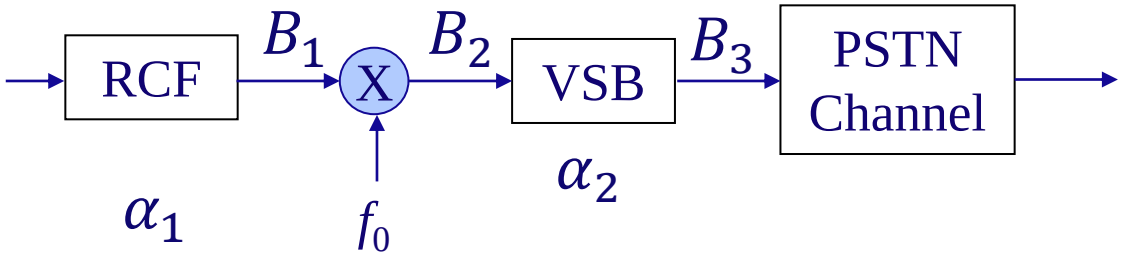
- Digital bandpass transmission, together with baseband waveform formation

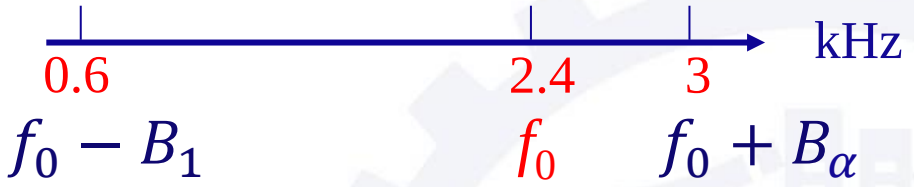


- Common used baseband transmission scheme without ISI
  1. **Raised cosine filter** (RCF)
  2. Partial response signaling (PR)

Example

An binary ASK system can use VSB (lower sideband) to transmit data at a rate of 2.4kb/s, via a PSTN channel with a passband 600~3000Hz; The carrier's frequency is f_0 =2.4KHz. (1) What is the VSB's roll-off factor ? (2) What is the roll-off factor of the raise cosine filter (RCF) ?





Sol.: (1) $\alpha_2 = B_\alpha / B_1 = \frac{3-2.4}{2.4-0.6} = 1/3$

(2) $\alpha_1 = B_1 / B_N - 1 = 1.8/(\frac{R_B}{2}) - 1 = 1/2$

PERFORMANCE SUMMARY OF THE BINARY DIGITAL MODULATIONS

- Given a certain bit error rate, the signal-to-noise ratio required by ASK, FSK and PSK differs by 3dB in turn
- Under the same conditions, the performance of coherent demodulation is better than that of incoherent demodulation
- Step by step analysis is needed for multistage modulation,